Finite-Sample Performance of High-Dimensional IVQR Estimators

Supervisor discussion report based on 500 Monte Carlo replications per design

Master Thesis Project

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Executive Summary

This report compares the finite-sample behavior of three instrumental-variable quantile-regression (IVQR) implementations for the thesis: a feasible DML-style IVQR procedure, an infeasible Oracle IVQR benchmark, and a feasible Mean-Lasso Post-selection IVQR procedure. All numerical evidence comes from the completed $R = 500$ full-run outputs in `results/report_full_run/`. Historical $R = 10$, $R = 20$, and $R = 100$ calibration experiments are intentionally excluded from the evidentiary base.

The completed full run contains 500 replications in each of 144 design cells. The implemented DML-style IVQR procedure attains **94.90%** coverage on resolved replications, compared with **94.19%** for Oracle IVQR and **92.42%** for Mean-Lasso Post-selection IVQR. The DML-style procedure's stronger calibration comes with a mean confidence-region length of 2.959 and a 33.09% full-grid rate. The central concern is Mean-Lasso Post-selection IVQR: its coverage falls to **90.42%** at $\tau = 0.75$, and the worst reported cell covers only **83.8%**.

The provisional assessment is:

- **DML-style IVQR – the most defensible implemented feasible procedure.** The implemented DML-style IVQR procedure is the most defensible feasible estimator among the three implementations considered, subject to its limited informativeness and implementation-specific nature. Its aggregate coverage is closest to 95%, but its regions are widest and frequently accept the full grid.
- **Oracle IVQR – retain as an infeasible benchmark.** It has the best point-estimation performance and useful confidence-region length, but knowing the true support cannot manufacture identifying variation. Its coverage is mildly below nominal overall and weaker in some cells.
- **Mean-Lasso Post-selection IVQR – not supported as a uniformly reliable 95% procedure.** It produces Oracle-like region lengths but systematically undercovers, particularly at the upper quantile. It remains informative as a comparison estimator, a negative result, and evidence on the cost of imperfect selection.

These classifications are implementation-specific. The simulations do not establish general validity for DML-IVQR or generic post-selection inference.

Questions for discussion

1. Is the implemented DML-style IVQR procedure sufficiently informative for the thesis, given its near-nominal average coverage but frequent wide and full-grid confidence regions?
2. Should Mean-Lasso Post-selection IVQR remain as a documented comparison or negative result, or should it be redesigned before final thesis inclusion?
3. Should point estimation be refined beyond the initial 0.2-spaced alpha grid before final thesis reporting?
4. Should the final thesis emphasize average performance or require a stronger design-level criterion, such as minimum or lower-tail coverage thresholds?

1 Research Objective

The target is the structural quantile treatment coefficient $\alpha_0(\tau)$ in a nonseparable outcome model. At quantile τ, the structural restriction can be written schematically as

$$P\{Y \leq D\alpha_0(\tau) + g_\tau(X) \mid X, Z\} = \tau,$$

where D is endogenous, Z is an excluded instrument, and X is a high-dimensional vector of controls. Ordinary quantile regression conditions on the endogenous treatment and therefore does not generally recover the causal structural quantile effect. IVQR instead uses instrument exogeneity and relevance to identify quantile-specific causal effects (Chernozhukov and Hansen 2005, 2006).

High-dimensional controls create a nuisance-estimation problem. Directly fitting hundreds of controls can be unstable; selecting a sparse subset can introduce selection bias; and flexible nuisance learning may overfit. The simulation therefore compares three roles:

1. **Oracle IVQR** knows the true active support and isolates the cost of nuisance estimation and selection, but is infeasible outside simulation.
2. **Mean-Lasso Post-selection IVQR** selects controls from the data and then applies IVQR, providing a repository-specific feasible sparse comparison. The label refers to the repository implementation and should not be interpreted as a general post-selection-valid IVQR procedure.
3. **DML-style IVQR** uses cross-fitted nuisance estimation and residualized scores to reduce first-order sensitivity to nuisance-learning error.

Good finite-sample behavior is multidimensional. The primary inferential criterion is empirical coverage near the nominal 95%, but a procedure that always reports the entire grid can cover without being informative. Assessment therefore also requires low bias and RMSE, reasonably short confidence regions, few unresolved replications, and robustness across τ, instrument strength, sample size, dimension, and DGP. No single scalar ranking is sufficient.

2 Confidence Intervals, Confidence Regions, and Coverage

2.1 Point estimates and set estimates

A point estimate $\hat{\alpha}$ selects one candidate value. A confidence set reports all candidate values not rejected by an inferential procedure. In a regular, strongly identified problem the set is often a connected interval. Here the test is inverted over an alpha grid, and the accepted set can be disconnected. **Confidence region** is therefore the default term: calling every set an interval would misdescribe unions of separated accepted components.

The nominal 95% level is a property of the procedure under repeated sampling. Empirical Monte Carlo coverage is the observed fraction of simulation replications in which the reported region contains the known true value. For a group of simulation rows $\mathcal{R}$, the report uses

$$\widehat{\mathrm{Cov}} = \frac{\sum_{r \in \mathcal{R}} 1\{\mathrm{CR}_r \text{ resolved}\} 1\{\alpha_{0,r} \in \mathrm{CR}_r\}}{\sum_{r \in \mathcal{R}} 1\{\mathrm{CR}_r \text{ resolved}\}}.$$

Unresolved cases are reported separately. They are neither silently counted as covered nor silently counted as uncovered. An explicitly `empty_valid` region is different: it is numerically resolved, contains no accepted value, has analysis length zero, and is uncovered. The legacy DML schema cannot distinguish a valid empty region from other causes of missing geometry, so missing DML CR triplets remain unresolved and status-unavailable.

2.2 Monte Carlo uncertainty

For R_{resolved} resolved replications and coverage estimate $\hat{p}$, the Monte Carlo standard error is

$$\text{MCSE}(\hat{p}) = \sqrt{\frac{\hat{p}(1-\hat{p})}{R_{\text{resolved}}}}.$$

At the design-cell level, $R_{\text{resolved}} \approx 500$ and $\hat{p} \approx 0.95$, giving an MCSE of about 0.98 percentage points. A rough 95% Monte Carlo interval is therefore about plus or minus two percentage points. Small deviations such as 94% versus 95% may be simulation noise in an individual cell; coverage near 84% is not.

Aggregated Monte Carlo intervals quantify simulation uncertainty around an average over heterogeneous design cells. They do not establish uniform coverage across designs.

2.3 Nonstandard confidence-region states

- A **disconnected region** contains multiple accepted blocks. It signals a nonstandard objective or weak/irregular identification, not a formatting error.
- An **empty region** rejects every candidate after a resolved evaluation. It is valid numerical output and counts as noncoverage.
- A **full-grid region** accepts every candidate in $[-1, 3]$. It covers the truth when the truth is on that grid, but offers little localization.
- An **unresolved region** has a numerical ambiguity that prevents a definitive covered/uncovered classification.

Coverage and informativeness must be read together. Wide or full-grid regions can be honest consequences of weak information; narrow regions are useful only if their calibration is credible. Test inversion and finite-sample pivotal methods motivate this set-valued perspective (Chernozhukov and Hansen 2008; Chernozhukov, Hansen, and Jansson 2009).

3 Weak Instruments and Weak Identification

An instrument must be **valid** (excluded from the structural disturbance in the relevant sense) and **relevant** (informative about endogenous treatment variation). Identification strength is not exhausted by the raw correlation between Z and D. It depends on the conditional model, the score's local sensitivity to α, nuisance adjustment, and the quantile under study. Consequently, an instrument can be informative for one structural quantile and weak for another (Jun 2008).

Conventional Wald approximations expand an estimator around a well-identified point and divide by an estimated local slope. When that slope is weak or unstable, normal approximations can be poor and standard errors can conceal the problem. Weak-identification-robust procedures instead test each candidate α and invert the test. The resulting region may be wide, disconnected, or the entire grid. Such a region is not automatically a failed estimator: it can be an honest statement that the sample does not localize the parameter (Chernozhukov and Hansen 2008).

This distinction matters directly here. The DML-style procedure's near-nominal coverage comes with wider regions and more full-grid outcomes. Orthogonalization and cross-fitting can reduce sensitivity to nuisance estimation error, but they do not strengthen the instrument. Near-nominal coverage can therefore coexist with wide or full-grid confidence regions. Their width reflects the combined effect of identification strength, the score, and the implemented inferential procedure; it is not mechanically caused by DML itself. Oracle knowledge of the true support cannot remove weak identification because support knowledge creates no new instrument variation. Mean-Lasso post-selection can look precise while under-covering if selection error or unaccounted selection uncertainty narrows the accepted set. Weak instruments are therefore part of the explanation, but not a universal explanation for all Mean-Lasso post-selection failures.

4 Estimators

4.1 Oracle IVQR

Oracle IVQR supplies the true union of active outcome and first-stage controls to the shared Chernozhukov–Hansen inverse-IVQR routine. It is infeasible in an empirical application because the true support is unknown. Its purpose is to benchmark the cost of learning nuisance structure. Oracle performance is not a theoretical upper bound: finite samples, grid point estimation, and weak identification remain even with the correct support.

4.2 Mean-Lasso Post-selection IVQR

The repository implementation fits mean-Lasso models for Y on X and D on X, takes the union of the selected controls, and applies the same inverse-IVQR routine. Selection is not quantile-specific, alpha-specific, cross-fitted, orthogonal, or formally selection-adjusted. It is therefore a heuristic sparse comparison specific to this repository rather than an implementation of a generic post-selection-valid procedure. It does not automatically account for model-selection uncertainty, so undercoverage is scientifically plausible. It is not an implementation of a uniformly valid post-selection theory. General post-selection results emphasize that regularization must be paired with orthogonalization or other protections for target inference (Chernozhukov, Hansen, and Spindler 2015). A short region after naive selection is not evidence that selection uncertainty has been handled.

4.3 DML-style IVQR

For each candidate α, the implementation cross-fits a penalized quantile regression of $Y - D\alpha$ on X and residualizes the excluded instrument Z on X using Ridge regression. It evaluates the cross-fitted scalar score

$$\psi_i(\alpha) = [\tau - 1\{Y_i - D_i\alpha - \hat{q}_\alpha(X_i) \leq 0\}]\, \widetilde{Z}_i$$

and inverts a weighted score statistic. Cross-fitting separates nuisance training from score evaluation, while residualization aims to reduce first-order sensitivity to nuisance error—the central DML logic (Chernozhukov et al. 2018).

The label **DML-style IVQR** is important. The code explicitly states that this is not the exact density-weighted procedure of Chen, Huang, and Tien (2021). The report therefore interprets the results as evidence for this repository's cross-fitted, residualized implementation, not as a generic validation of every DML-IVQR estimator.

5 Monte Carlo Design

The outcome has a binary endogenous treatment generated from a latent index. DGP1 is a sparse Gaussian baseline; DGP2 is a denser sparse selection-stress design; DGP3 uses heavy-tailed, variance-normalized Student-t_5 structural shocks. The true structural effect varies with τ, so bias and coverage are evaluated against the appropriate DGP–quantile truth rather than a single constant.

All estimators use the same design-replication keys and seeds. There are $3 \times 2 \times 2 \times 4 \times 3 = 144$ cells and 500 replications per cell for each estimator. The reconciled panel contains 72,000 rows per estimator and 216,000 in total, with no duplicate design-replication keys.

Pooled estimator results therefore use approximately 72,000 resolved replications, while a pooled estimator–quantile result uses approximately 24,000. These large effective counts make simulation error around the pooled average small, but the underlying design cells remain heterogeneous. Aggregated Monte Carlo intervals quantify simulation uncertainty around an average over heterogeneous design cells. They do not establish uniform coverage across designs.

Table 1: Validated Monte Carlo design

Component	Validated/configured value
DGPs	DGP1, DGP2, DGP3
Active support	DGP1: 5 active controls, DGP2: 10 active controls, DGP3: 5 active controls
Sample sizes	500, 1000
Control dimensions	200, 500
Instrument-strength index pi	0.1, 0.25, 0.5, 1.0
Quantiles tau	0.25, 0.5, 0.75
Replications per cell	500
Design cells	144
True alpha by tau=(0.25, 0.50, 0.75)	DGP1: (0.3255, 1.0000, 1.6745); DGP2: (0.3255, 1.0000, 1.6745); DGP3: (0.4371, 1.0000, 1.5629)
Initial alpha grid	[-1, 3], 21 points, spacing 0.2
CR refinement	Adaptive boundary refinement; tolerance 0.025
Nominal confidence level	95%
Critical-value multiplier	1
Estimators	DML-style residualized IVQR; true-support Oracle IVQR; Mean-Lasso Post-selection IVQR

Note: Configured values are cross-checked against validation.json and the current simulation code.

Table 2: Overall finite-sample performance

Estimator	Coverage	Bias	MAE	RMSE	Estimate SD	Mean CR length	Median CR length	Full grid	Unresolved
DML-style IVQR	94.90%	0.009	0.760	0.987	1.045	2.959	3.188	33.09%	0.06%
Oracle IVQR	94.19%	-0.007	0.602	0.836	0.930	2.407	2.235	19.37%	0.00%
Mean-Lasso Post-selection IVQR	92.42%	-0.057	0.624	0.859	0.928	2.380	2.199	18.22%	0.00%

Note: Coverage is conditional on resolved replications. CR denotes confidence region. Percentages use the authoritative validated summaries.

6 Overall Results

The implemented DML-style IVQR procedure has the highest overall coverage (94.90%), but also the longest mean region (2.959). Oracle IVQR has the smallest MAE (0.602) and RMSE (0.836), making it the strongest point-estimation benchmark. Mean-Lasso Post-selection IVQR produces a slightly shorter mean region than Oracle (2.380 versus 2.407) but covers only 92.42%. The shorter set is therefore not evidence of superior inference.

The ranking changes with the criterion. Oracle performs best for point accuracy; the DML-style procedure performs closest to nominal average coverage in these simulations; Mean-Lasso Post-selection IVQR is slightly more compact than Oracle but inferentially less reliable. Coverage must not be confused with precision, and precision must not be confused with validity.

7 Design-Level Coverage Dispersion

Overall averages conceal meaningful design-level heterogeneity. The implemented DML-style IVQR procedure has a lower-tail (10th-percentile) cell coverage of 93.60%, and 0.00% of its cells fall below 90%. Oracle IVQR shows mild finite-sample undercoverage in its lower tail, with a 10th percentile of 92.40%. Mean-Lasso Post-selection IVQR has both frequent and severe undercoverage: 11.11% of cells fall below 90%, and its minimum is 83.8%. Average calibration must therefore be assessed alongside lower-tail cell performance.

Table 3: Distribution of resolved-replication coverage across design cells

Estimator	Cells	Min	P10	P25	Median	P75	P90	Max	Below 90%	Below 92.5%	Below 95%	At least 95%
DML-style IVQR	144	91.80%	93.60%	94.20%	94.80%	95.79%	96.20%	97.39%	0.00%	0.69%	52.08%	47.92%
Oracle IVQR	144	91.00%	92.40%	93.00%	94.40%	95.25%	95.80%	97.40%	0.00%	11.81%	65.97%	34.03%
Mean-Lasso Post-selection IVQR	144	83.80%	89.66%	90.80%	92.80%	94.00%	95.00%	96.60%	11.11%	43.06%	88.19%	11.81%

Note: Each estimator has 144 design cells. Quantiles and threshold shares use empirical coverage already reported in the authoritative design-cell summary; no row-level denominator is recomputed.

This table uses the resolved-cell coverage values in `table_05_by_design_cell.csv` directly. Overall averages can hide severe cell-level failures: DML-style IVQR must be assessed by both average coverage and lower-tail cell performance, Oracle by its mild finite-sample undercoverage, and Mean-Lasso Post-selection IVQR by both the frequency and severity of its undercoverage.

8 Results by Quantile

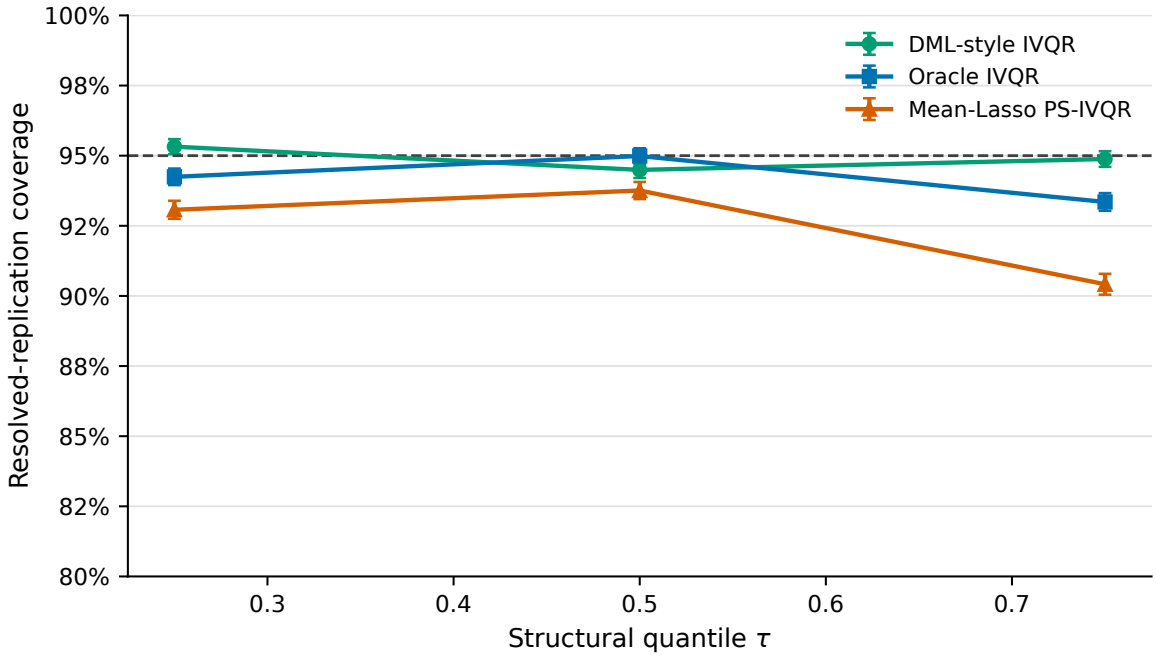

Figure 1: Resolved-replication coverage by structural quantile. Each estimator–quantile point pools approximately 24,000 replications over heterogeneous designs. Bars are normal 95% Monte Carlo intervals around the pooled average; they do not establish uniform coverage across design cells.

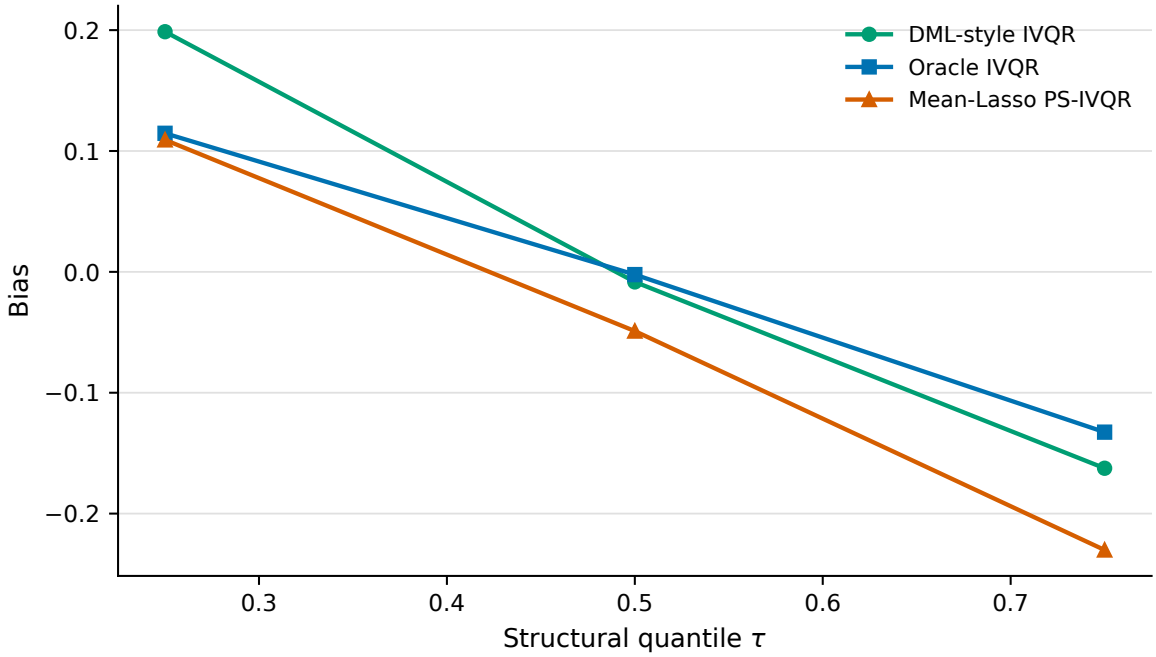

Figure 2: Bias by structural quantile. The upper-quantile Mean-Lasso post-selection result is distinguished by a materially more negative bias.

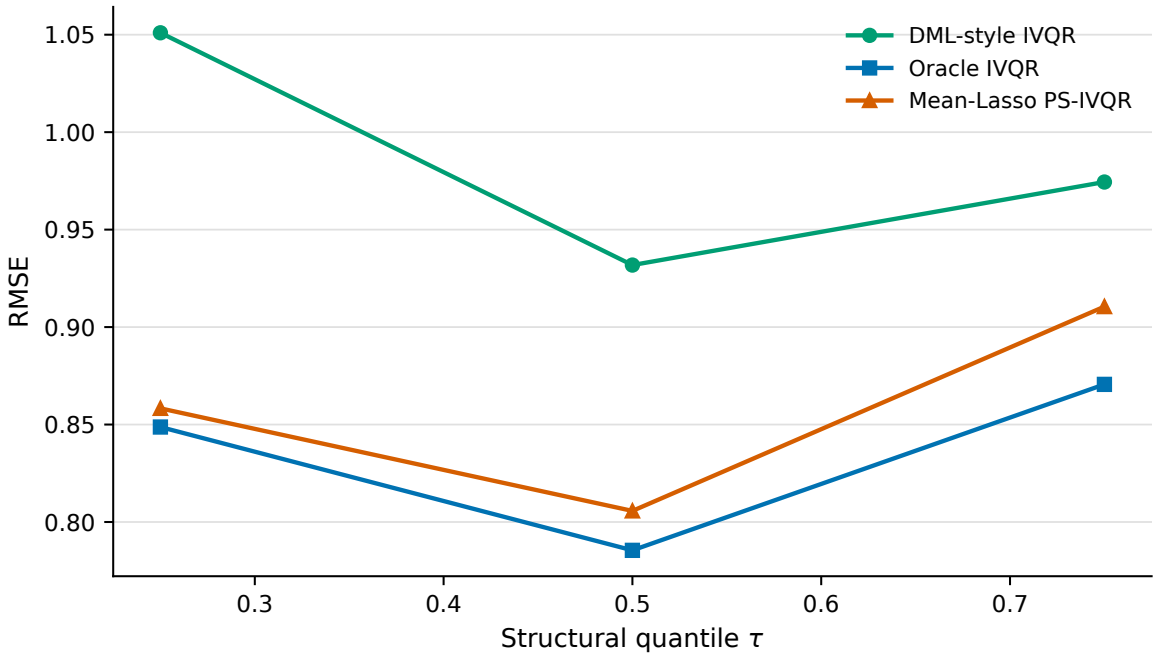

Figure 3: RMSE by structural quantile. RMSE combines variance and bias and should be read alongside coverage rather than as a substitute for it.

The main instability is concentrated in the upper quantile. Mean-Lasso Post-selection IVQR coverage is 90.42% at $\tau = 0.75$, with bias -0.230; its coverage is 93.07% at $\tau = 0.25$. Oracle also weakens at $\tau = 0.75$ but less severely, while DML remains close to nominal at each reported quantile. The coincident negative upper-quantile bias is consistent with, but does not by itself prove, a selection mechanism.

The aggregate error bars in Figure 1 are narrow because each point pools roughly 24,000 replications. Aggregated Monte Carlo intervals quantify simulation uncertainty around an average over heterogeneous

design cells. They do not establish uniform coverage across designs. The upper-quantile Mean-Lasso post-selection gap is far larger than that simulation uncertainty.

9 Results by Instrument Strength

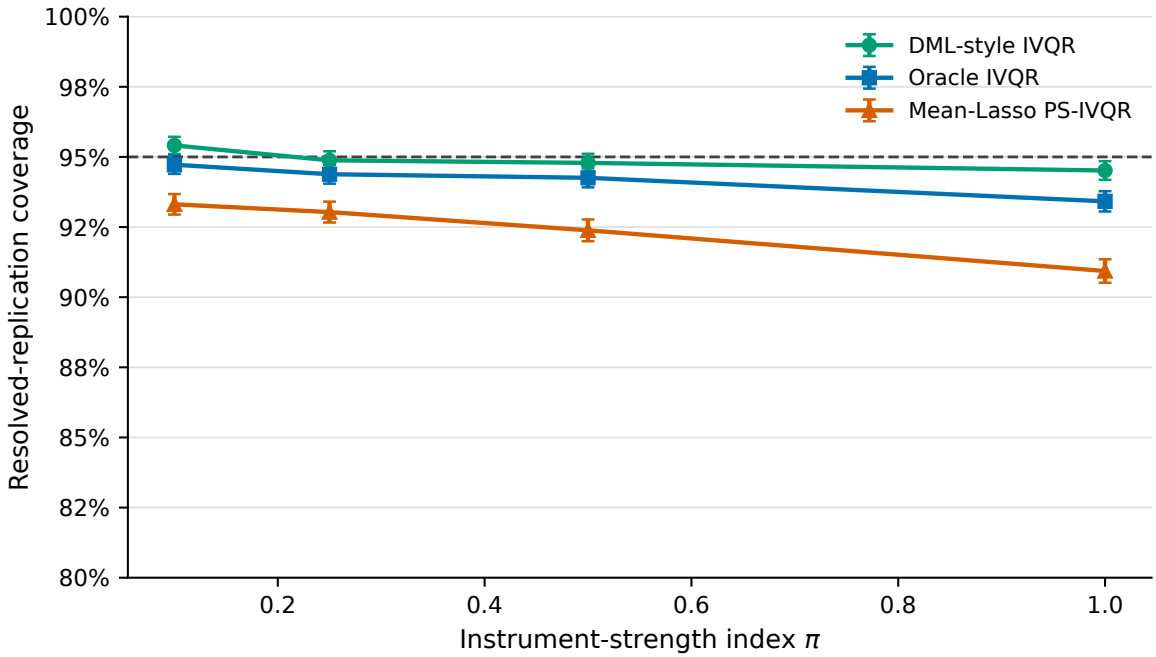

Figure 4: Coverage by the instrument-strength index. The vertical scale is explicitly restricted to 80–100% to show calibration differences; the nominal 95% line remains visible.

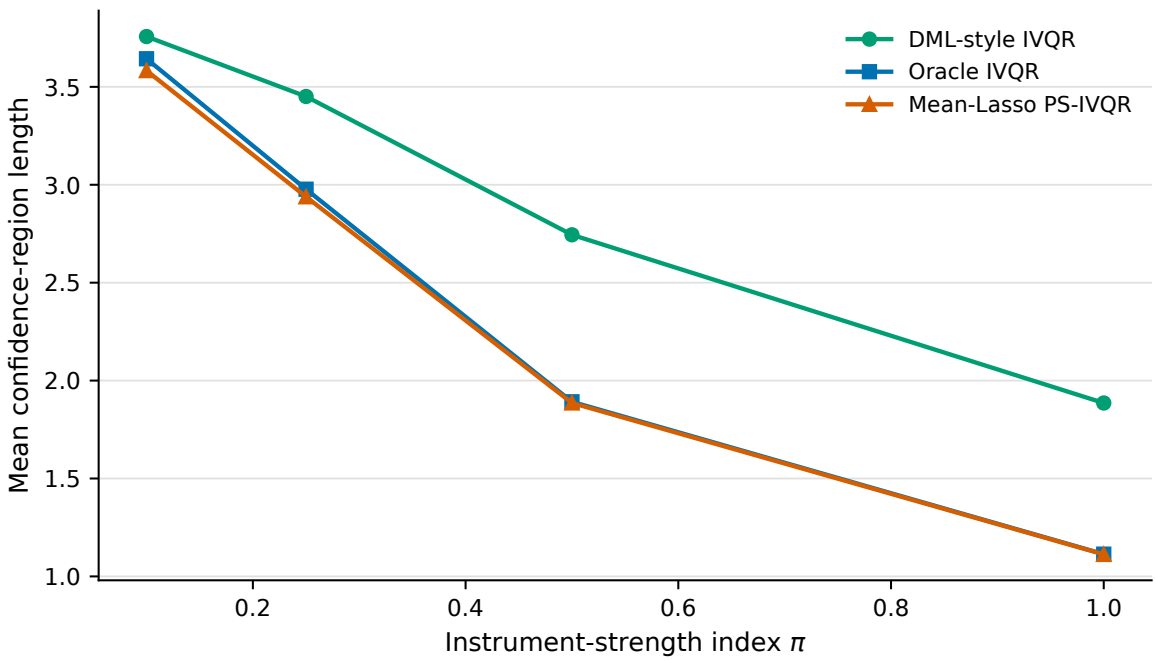

Figure 5: Mean confidence-region length by instrument strength. Falling length with stronger instruments reflects improved localization, not automatically improved calibration.

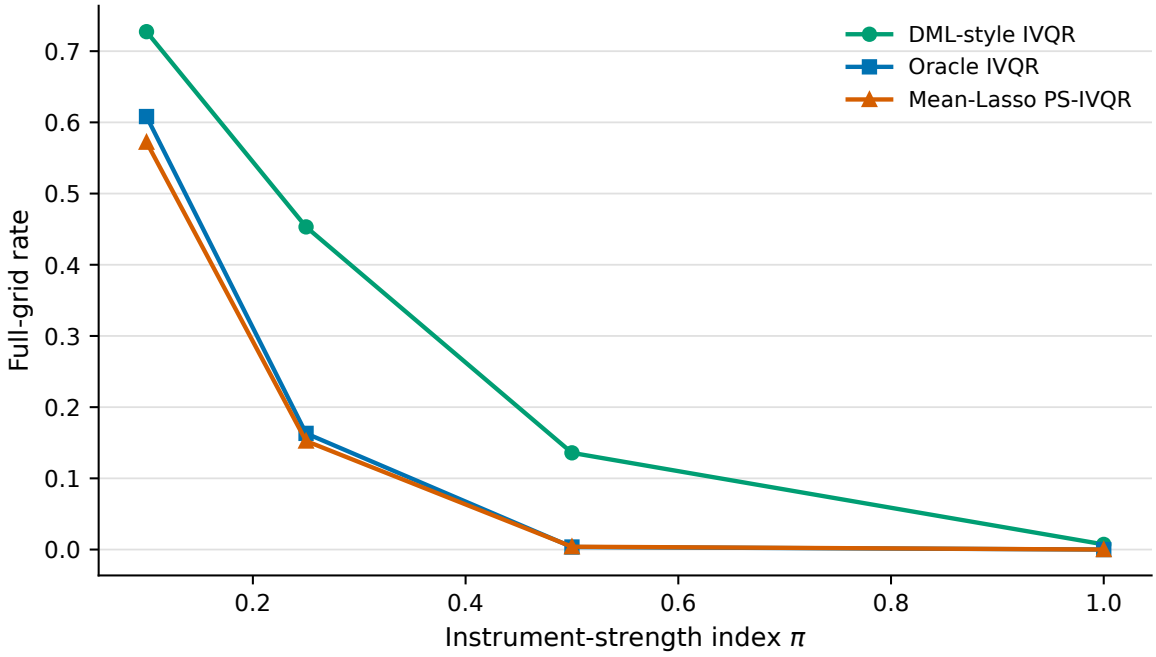

Figure 6: Full-grid rate by instrument strength. Full-grid outcomes are most common when the instrument-strength index is smallest.

All estimators become more informative as the instrument-strength index increases. For $\pi = 0.1$, mean CR lengths are 3.757 (DML-style), 3.643 (Oracle), and 3.583 (Mean-Lasso post-selection). At $\pi = 1$, the corresponding lengths fall to 1.886, 1.114, and 1.112. This is the expected informativeness response to stronger identification. Mean-Lasso post-selection undercoverage, however, persists—and is most visible at $\pi = 1$—so weak identification is not its only plausible source.

The Oracle and Mean-Lasso post-selection length curves are close, whereas the DML-style procedure is wider, especially for weak designs. Stronger instruments reduce full-grid rates sharply. Yet Mean-Lasso post-selection coverage declines as π rises from 0.1 to 1.0. This pattern is consistent with selection-related bias becoming visible when confidence regions narrow; it is not evidence that instrument strength causes the bias.

Table 4: Ten lowest-coverage design cells: design and coverage

Estimator	DGP	n	p	pi	tau	Resolved	Coverage	MCSE
Mean-Lasso Post-selection IVQR	DGP2	500	500	1	0.75	500	83.80%	1.65%
Mean-Lasso Post-selection IVQR	DGP3	500	500	1	0.75	500	83.80%	1.65%
Mean-Lasso Post-selection IVQR	DGP2	500	200	1	0.75	500	85.80%	1.56%
Mean-Lasso Post-selection IVQR	DGP2	500	500	0.25	0.75	500	86.80%	1.51%
Mean-Lasso Post-selection IVQR	DGP2	500	500	0.5	0.75	500	87.00%	1.50%
Mean-Lasso Post-selection IVQR	DGP2	500	200	0.5	0.75	500	88.00%	1.45%
Mean-Lasso Post-selection IVQR	DGP2	1000	500	0.5	0.75	500	88.00%	1.45%
Mean-Lasso Post-selection IVQR	DGP2	1000	500	1	0.75	500	88.00%	1.45%
Mean-Lasso Post-selection IVQR	DGP1	1000	500	1	0.75	500	88.20%	1.44%
Mean-Lasso Post-selection IVQR	DGP2	1000	200	0.1	0.75	500	88.40%	1.43%

Note: Cells are ranked by resolved-replication coverage. MCSE is the Monte Carlo standard error of coverage.

10 Precision and Informativeness

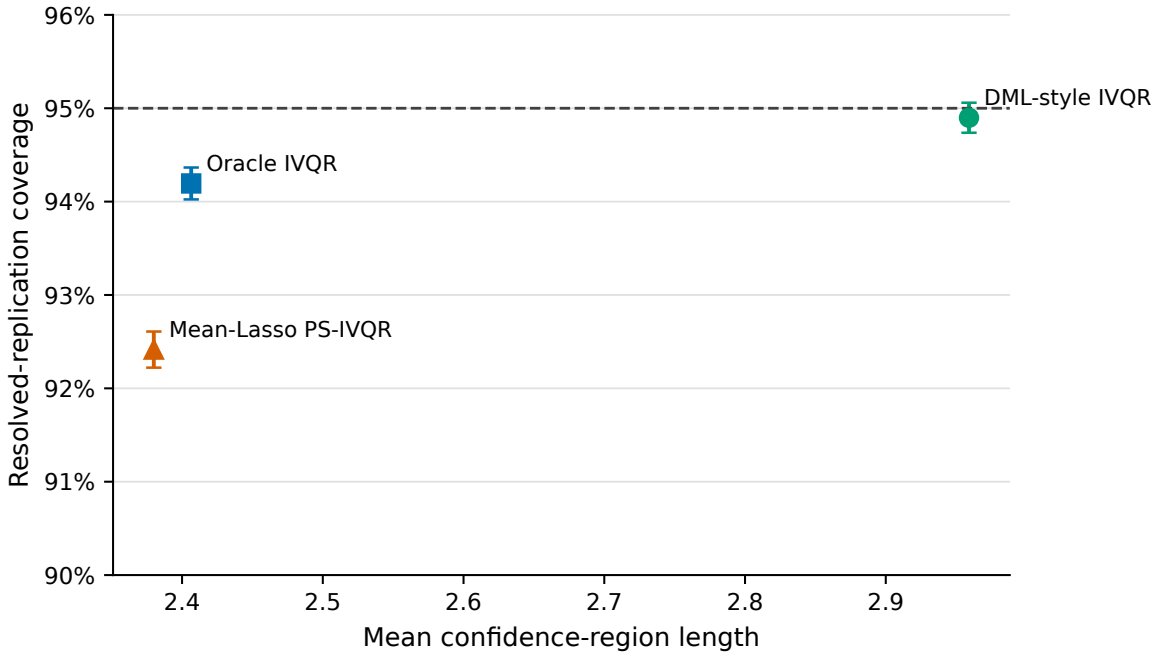

Figure 7: Overall coverage versus mean confidence-region length. Vertical bars show 95% Monte Carlo uncertainty for coverage. Moving left is attractive only if coverage remains calibrated.

The trade-off in Figure 7 is not a conventional efficiency frontier because the estimators do not all satisfy the same coverage constraint. The DML-style procedure occupies the high-coverage, low-informativeness position. Oracle gives the strongest balance of point accuracy and region length, though it is infeasible. Mean-Lasso post-selection's length is attractive in isolation but its coverage is inadequate.

Decision rule: A shorter confidence region is not preferable if it fails to achieve nominal coverage.

Full-grid DML-style regions should not be hidden as failures: under weak identification, they can be the correct statement that the candidate grid cannot be narrowed. They do, however, limit the estimator's substantive usefulness. The supervisor decision must therefore balance honest uncertainty against the thesis's need for informative empirical conclusions.

Table 5: Ten lowest-coverage design cells: performance and diagnostics

Estimator	DGP	n	p	pi	tau	Bias	RMSE	Mean CR length	Full grid	Unresolved
Mean-Lasso Post-selection IVQR	DGP2	500	500	1	0.75	-0.256	0.516	1.478	0.00%	0.00%
Mean-Lasso Post-selection IVQR	DGP3	500	500	1	0.75	-0.157	0.391	1.171	0.00%	0.00%
Mean-Lasso Post-selection IVQR	DGP2	500	200	1	0.75	-0.146	0.452	1.452	0.00%	0.00%
Mean-Lasso Post-selection IVQR	DGP2	500	500	0.25	0.75	-0.472	1.279	3.244	35.00%	0.00%
Mean-Lasso Post-selection IVQR	DGP2	500	500	0.5	0.75	-0.352	0.840	2.502	2.00%	0.00%
Mean-Lasso Post-selection IVQR	DGP2	500	200	0.5	0.75	-0.270	0.825	2.513	3.60%	0.00%
Mean-Lasso Post-selection IVQR	DGP2	1000	500	0.5	0.75	-0.154	0.570	1.836	0.00%	0.00%
Mean-Lasso Post-selection IVQR	DGP2	1000	500	1	0.75	-0.107	0.339	1.008	0.00%	0.00%
Mean-Lasso Post-selection IVQR	DGP1	1000	500	1	0.75	-0.086	0.290	0.940	0.00%	0.00%
Mean-Lasso Post-selection IVQR	DGP2	1000	200	0.1	0.75	-0.402	1.327	3.560	59.00%	0.00%

Note: Rows correspond exactly to the cells in the preceding table. The split avoids compressing fourteen columns into one table.

Table 6: Numerical and confidence-region diagnostics

Estimator	Unresolved	Empty	Disconnected	Boundary estimate	Iteration warning	Rank failure	Refinement limit
DML-style IVQR	0.06%	NA	NA	7.66%	NA	NA	NA
Oracle IVQR	0.00%	0.02%	15.95%	5.35%	70.01%	0.00%	0.00%
Mean-Lasso Post-selection IVQR	0.00%	0.06%	16.48%	5.56%	89.86%	0.00%	0.00%

Note: NA means the source schema does not provide that diagnostic. It must not be interpreted as zero.

11 Worst-Performing Designs

The lowest-coverage cell is Mean-Lasso Post-selection IVQR, DGP2, $n = 500$, $p = 500$, $\pi = 1$, and $\tau = 0.75$, with coverage 83.8% and MCSE 1.65%. A 95% Monte Carlo interval is approximately [80.6%, 87.0%], far below 95%. Sampling noise from 500 replications cannot plausibly explain that gap.

The worst ten cells are all Mean-Lasso post-selection cells and heavily concentrated at $\tau = 0.75$. Several occur with $\pi = 0.5$ or 1, so the tail failures cannot be attributed solely to the weakest-instrument configuration. They also occur across DGPs and both control dimensions. This is evidence of systematic heterogeneity in inferential performance, not one anomalous simulation cell.

12 Compact Numerical Diagnostics

The resolved denominator excludes 43 DML rows (0.060%) and two Mean-Lasso post-selection rows (0.003%); Oracle has no unresolved rows. DML's legacy file contains no explicit CR status, component, iteration-warning, rank-failure, or refinement metadata. Those entries are intentionally reported as **NA**, not zero. Oracle and Mean-Lasso post-selection record zero rank failures and zero refinement-limit hits in these validated summaries.

Iteration warnings are common in the richer Oracle and Mean-Lasso post-selection schemas, but the production policy retains warning fits when their numerical output is otherwise usable. Rank-failure and refinement-limit rates are zero in these validated results. DML lacks the metadata needed to make the same statements; reporting zero would fabricate information.

12.1 Initial-grid point estimation

The point estimate $\hat{\alpha}$ is selected from the initial 21-point grid $[-1, 3]$, whose spacing is 0.2. Adaptive midpoint probing and refinement improve confidence-region topology and boundary location, but the production setting `alpha_hat_grid="initial"` does not use those added evaluations to refine the point estimate. Boundary-estimate rates are therefore reported as a diagnostic.

This discrete point-estimation rule can affect bias, RMSE, and boundary mass. It does not by itself show that inverted-region inference is invalid: point selection and test inversion use related but distinct features of the evaluated objective. Nevertheless, a refined-point robustness analysis is a sensible candidate before final thesis submission.

13 Interpretation and Thesis Decision

13.1 DML-style IVQR: most defensible implemented feasible procedure

The implemented DML-style IVQR procedure is approximately nominal overall, remains close to 95% across quantiles, and has very few status-unavailable rows. Its limitation is informativeness: weak designs often generate long or full-grid regions. The appropriate thesis claim is not that DML-IVQR is uniformly precise or generally validated. The implemented DML-style IVQR procedure is the most defensible feasible estimator among the three implementations considered, subject to its limited informativeness and implementation-specific nature. Aggregated Monte Carlo intervals quantify simulation uncertainty around an average over heterogeneous design cells. They do not establish uniform coverage across designs.

13.2 Oracle IVQR: benchmark, not empirical procedure

Oracle has the strongest point accuracy and a useful coverage–length balance. It is mildly below nominal overall and weaker in some upper-quantile designs, confirming that nuisance support is not the only finite-sample difficulty. **Provisional decision:** retain as the infeasible benchmark for interpreting nuisance-learning costs; do not present it as an available empirical estimator.

13.3 Mean-Lasso Post-selection IVQR: comparison or redesign

Mean-Lasso Post-selection IVQR produces Oracle-like confidence-region lengths but systematic undercoverage, with a pronounced upper-quantile problem and severe worst-cell failures. Its current mean-Lasso union step does not implement formal selection-adjusted inference. **Decision implication:** do not present it as a uniformly reliable 95% procedure. It can remain as a comparison estimator, a negative result, and evidence about the cost of imperfect variable selection. If the thesis requires it as a positive feasible method, redesign and fresh validation would be necessary; this report does not authorize or perform that simulation work.

These findings establish finite-sample performance only for the repository implementation and the simulated designs considered here. They do not constitute a general validity result for DML-IVQR.

14 Conclusion

Among the implemented feasible procedures, DML-style IVQR performs closest to nominal average coverage, but frequent wide and full-grid regions materially limit its informativeness. Oracle remains an infeasible point-estimation benchmark and exhibits mild finite-sample undercoverage. Mean-Lasso Mean-Lasso Post-selection IVQR shows severe, systematic design-level undercoverage and is best treated as a repository-specific comparison or negative result unless it is redesigned and evaluated in new simulations. These conclusions concern only the repository implementation and the simulated designs, not DML-IVQR or post-selection inference in general.

15 Questions for the Supervisor

Decision questions

1. **DML informativeness.** Is the implemented DML-style IVQR procedure sufficiently informative for the thesis, given its near-nominal average coverage but frequent wide and full-grid confidence regions?
2. **Role of Mean-Lasso post-selection.** Should Mean-Lasso Post-selection IVQR remain as a documented comparison or negative result, or should it be redesigned before final thesis inclusion?
3. **Point-estimate grid.** Should point estimation be refined beyond the initial 0.2-spaced alpha grid before final thesis reporting?
4. **Design-level criterion.** Should the final thesis emphasize average performance or require a stronger design-level criterion, such as minimum or lower-tail coverage thresholds?

Table 7: Performance by sample size and control dimension

Estimator	n	p	Coverage	Bias	RMSE	Mean CR length	Full grid	Unresolved
DML-style IVQR	500	200	94.72%	0.022	1.034	3.123	36.95%	0.05%
DML-style IVQR	500	500	94.94%	0.022	1.073	3.238	39.96%	0.07%
DML-style IVQR	1000	200	95.00%	-0.021	0.824	2.468	21.51%	0.03%
DML-style IVQR	1000	500	94.94%	0.015	0.998	3.008	33.94%	0.09%
Oracle IVQR	500	200	94.17%	-0.004	0.897	2.606	23.08%	0.00%
Oracle IVQR	500	500	94.21%	-0.009	0.892	2.597	22.78%	0.00%
Oracle IVQR	1000	200	94.14%	-0.005	0.777	2.209	15.59%	0.00%
Oracle IVQR	1000	500	94.26%	-0.009	0.767	2.215	16.03%	0.00%
Mean-Lasso Post-selection IVQR	500	200	92.46%	-0.049	0.921	2.577	21.86%	0.00%
Mean-Lasso Post-selection IVQR	500	500	91.77%	-0.092	0.933	2.572	21.36%	0.00%
Mean-Lasso Post-selection IVQR	1000	200	92.86%	-0.036	0.783	2.182	14.77%	0.01%
Mean-Lasso Post-selection IVQR	1000	500	92.58%	-0.049	0.788	2.189	14.88%	0.01%

Note: Coverage uses resolved replications only.

Table 9: Detailed standardized confidence-region statuses

Estimator	Status	Rows	Resolved	Covered	Uncovered	Unresolved	Coverage
DML-style IVQR	full_grid_observed_status_unavailable	23825	23825	23825	0	0	100.00%
DML-style IVQR	missing_status_unavailable	43	0	0	0	43	NA
DML-style IVQR	observed_status_unavailable	48132	48132	44461	3671	0	92.37%
Oracle IVQR	empty_valid	13	13	0	13	0	0.00%
Oracle IVQR	full_grid_valid	13948	13948	13948	0	0	100.00%
Oracle IVQR	valid	58039	58039	53872	4167	0	92.82%
Mean-Lasso Post-selection IVQR	empty_valid	41	41	0	41	0	0.00%
Mean-Lasso Post-selection IVQR	full_grid_valid	13115	13115	13115	0	0	100.00%
Mean-Lasso Post-selection IVQR	partially_unresolved	2	0	0	0	2	NA
Mean-Lasso Post-selection IVQR	valid	58842	58842	53422	5420	0	90.79%

Note: DML status labels explicitly state that detailed legacy status metadata are unavailable.

A Results by Sample Size and Dimension

This aggregation checks whether the headline rankings are driven by a single (n, p) configuration. It should be interpreted jointly with the complete cell table, because aggregation over DGP, π, and τ can conceal difficult subdesigns.

B Confidence-Region Status Definitions

Standardized status	Meaning	Coverage treatment
`valid`	Resolved nonempty region that is not the full grid	Included
`full_grid_valid`	Resolved region accepts the entire grid	Included
`empty_valid`	Resolved accepted set is empty	Included as uncovered
`partially_unresolved`	Some evaluations prevent definitive topology/coverage	Excluded
`fully_unresolved`	No definitive numerical region	Excluded
`observed_status_unavailable`	Legacy DML geometry observed, detailed status absent	Included using the documented complete-triplet proxy
`missing_status_unavailable`	Legacy DML CR geometry missing and cause unknown	Excluded and reported separately

C Selected-Control Diagnostic

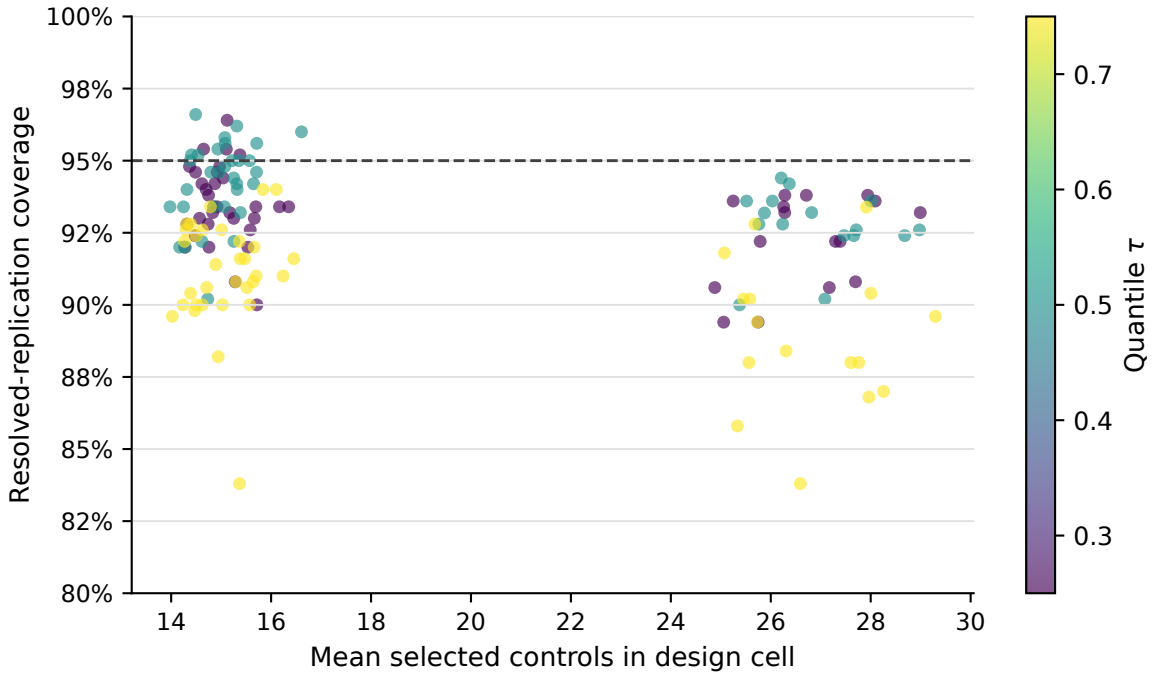

Figure 8: Mean-Lasso post-selection design-cell coverage versus the mean number of selected controls. Color denotes the structural quantile. This descriptive plot does not identify a causal effect of selection size on coverage.

Selected-control counts are genuinely available for Mean-Lasso post-selection and are shown only as a descriptive diagnostic. The relationship mixes DGP, sample size, dimension, instrument strength, and quantile. It cannot establish that selecting more or fewer controls causes a particular coverage outcome.

D Estimator-Specific Implementation Notes

- **Shared inversion.** Oracle and Mean-Lasso post-selection use a Chernozhukov–Hansen-style inverse-IVQR grid routine with midpoint-assisted adaptive refinement of confidence-region boundaries.
- **Oracle support.** The active support is the exact union of nonzero outcome and first-stage coefficients: five controls in DGP1/DGP3 and ten in DGP2.
- **Mean-Lasso post-selection.** Mean-Lasso control sets from $Y \sim X$ and $D \sim X$ are unioned. All excluded instruments are retained. The method is not quantile-specific or formally selection-adjusted.
- **DML-style procedure.** Candidate-specific penalized quantile nuisance models and Ridge instrument residualization are cross-fitted. The score is scalar because the implementation supports one excluded instrument. It is not the exact published density-weighted DML-IVQR algorithm.
- **Point versus region grid.** Point estimates use the initial grid; adaptive evaluations refine the confidence region.

E Reproducibility and Data Limitations

From the repository root, run the exact one-line Windows PowerShell command:

```
pixi run build_full_run_report
```

For the shorter meeting version, run:

```
pixi run build_full_run_meeting_report
```

The task first regenerates all assets from the reconciled summary tables and then renders this document. The asset module does not read raw simulation files and does not recompute coverage. Its manifest explicitly records the validated summary source. To run the stages separately:

```
pixi run python -m analysis.full_run_report
quarto render documents\Full_Run_Results.qmd --to pdf
```

Paths containing parentheses must be quoted in PowerShell, although this report does not require an alternate raw Oracle path.

The principal data limitation is the legacy DML schema. Forty-three DML rows have missing CR triplets but no status field that can distinguish an empty valid region from numerical failure or another cause. DML also lacks components, disconnectedness, iteration-warning, rank-failure, and refinement-limit metadata. These fields remain **NA**. Oracle and Mean-Lasso post-selection have richer versioned CR schemas. Finally, the current summary outputs do not prove that every historical command-line tuning label matches the file-generation environment; the report therefore describes implementation behavior and validated outputs without using old calibration tables as final evidence.

F Complete Design-Cell Results

The following long table reproduces all 432 estimator–design cells from the validated `table_05_by_design_cell.csv`. It is included for auditability rather than main-text interpretation.

Table 10: Complete results by estimator and design cell

Estimator	DGP	n	p	pi	tau	R-res.	Coverage	Bias	RMSE	Mean CR	Full grid	Unresolved
DML-style IVQR	DGP1	500	200	0.1	0.25	500	94.60%	0.766	1.516	3.782	74.80%	0.00%
DML-style IVQR	DGP2	500	200	0.1	0.25	500	95.80%	0.506	1.385	3.791	76.00%	0.00%
DML-style IVQR	DGP3	500	200	0.1	0.25	500	95.20%	0.686	1.462	3.789	73.40%	0.00%
DML-style IVQR	DGP1	500	500	0.1	0.25	500	96.20%	0.704	1.458	3.795	75.80%	0.00%
DML-style IVQR	DGP2	500	500	0.1	0.25	498	96.18%	0.652	1.433	3.799	76.40%	0.40%
DML-style IVQR	DGP3	500	500	0.1	0.25	498	94.78%	0.556	1.387	3.772	71.80%	0.40%
DML-style IVQR	DGP1	1000	200	0.1	0.25	500	97.00%	0.462	1.342	3.684	65.40%	0.00%
DML-style IVQR	DGP2	1000	200	0.1	0.25	500	96.20%	0.556	1.375	3.756	74.00%	0.00%
DML-style IVQR	DGP3	1000	200	0.1	0.25	499	94.79%	0.394	1.316	3.656	59.80%	0.20%
DML-style IVQR	DGP1	1000	500	0.1	0.25	498	96.39%	0.572	1.431	3.790	72.80%	0.40%
DML-style IVQR	DGP2	1000	500	0.1	0.25	499	97.39%	0.602	1.404	3.793	77.00%	0.20%
DML-style IVQR	DGP3	1000	500	0.1	0.25	499	95.99%	0.518	1.403	3.762	72.80%	0.20%
DML-style IVQR	DGP1	500	200	0.25	0.25	500	95.80%	0.336	1.269	3.601	54.20%	0.00%
DML-style IVQR	DGP2	500	200	0.25	0.25	498	94.78%	0.363	1.239	3.664	60.20%	0.40%
DML-style IVQR	DGP3	500	200	0.25	0.25	500	95.40%	0.367	1.214	3.561	47.60%	0.00%
DML-style IVQR	DGP1	500	500	0.25	0.25	499	95.79%	0.386	1.330	3.685	61.80%	0.20%
DML-style IVQR	DGP2	500	500	0.25	0.25	500	95.40%	0.372	1.304	3.677	65.00%	0.00%
DML-style IVQR	DGP3	500	500	0.25	0.25	500	96.00%	0.208	1.202	3.579	51.20%	0.00%
DML-style IVQR	DGP1	1000	200	0.25	0.25	499	96.99%	0.096	0.978	3.137	20.40%	0.20%
DML-style IVQR	DGP2	1000	200	0.25	0.25	499	95.59%	0.115	0.986	3.323	30.40%	0.20%
DML-style IVQR	DGP3	1000	200	0.25	0.25	500	95.80%	-0.064	0.870	2.891	9.20%	0.00%
DML-style IVQR	DGP1	1000	500	0.25	0.25	500	95.00%	0.317	1.208	3.533	48.60%	0.00%
DML-style IVQR	DGP2	1000	500	0.25	0.25	500	94.80%	0.340	1.249	3.662	64.00%	0.00%
DML-style IVQR	DGP3	1000	500	0.25	0.25	500	93.80%	0.218	1.163	3.471	43.60%	0.00%
DML-style IVQR	DGP1	500	200	0.5	0.25	500	96.20%	0.054	0.925	3.055	16.80%	0.00%
DML-style IVQR	DGP2	500	200	0.5	0.25	500	95.80%	0.088	0.977	3.233	27.40%	0.00%
DML-style IVQR	DGP3	500	200	0.5	0.25	500	95.00%	-0.020	0.832	2.905	10.00%	0.00%
DML-style IVQR	DGP1	500	500	0.5	0.25	500	96.20%	0.117	0.965	3.280	26.00%	0.00%
DML-style IVQR	DGP2	500	500	0.5	0.25	500	94.60%	0.088	1.065	3.311	36.00%	0.00%
DML-style IVQR	DGP3	500	500	0.5	0.25	499	95.79%	0.024	0.902	3.061	16.80%	0.20%
DML-style IVQR	DGP1	1000	200	0.5	0.25	500	95.80%	-0.102	0.588	2.136	0.20%	0.00%
DML-style IVQR	DGP2	1000	200	0.5	0.25	500	95.20%	-0.087	0.636	2.356	0.80%	0.00%
DML-style IVQR	DGP3	1000	200	0.5	0.25	500	96.40%	-0.044	0.465	1.952	0.00%	0.00%
DML-style IVQR	DGP1	1000	500	0.5	0.25	499	95.79%	0.008	0.827	2.922	10.00%	0.20%
DML-style IVQR	DGP2	1000	500	0.5	0.25	499	93.39%	0.018	0.973	3.070	19.00%	0.20%
DML-style IVQR	DGP3	1000	500	0.5	0.25	500	93.80%	-0.051	0.789	2.709	5.80%	0.00%
DML-style IVQR	DGP1	500	200	1	0.25	500	92.60%	-0.045	0.623	2.222	0.40%	0.00%
DML-style IVQR	DGP2	500	200	1	0.25	500	93.80%	-0.066	0.703	2.468	2.60%	0.00%
DML-style IVQR	DGP3	500	200	1	0.25	500	95.20%	-0.060	0.561	2.106	0.00%	0.00%
DML-style IVQR	DGP1	500	500	1	0.25	500	94.00%	-0.049	0.679	2.440	1.60%	0.00%
DML-style IVQR	DGP2	500	500	1	0.25	500	96.20%	-0.033	0.729	2.685	4.60%	0.00%
DML-style IVQR	DGP3	500	500	1	0.25	500	95.20%	-0.032	0.617	2.335	1.20%	0.00%
DML-style IVQR	DGP1	1000	200	1	0.25	500	93.60%	-0.068	0.360	1.357	0.00%	0.00%
DML-style IVQR	DGP2	1000	200	1	0.25	500	94.80%	-0.058	0.389	1.520	0.00%	0.00%
DML-style IVQR	DGP3	1000	200	1	0.25	500	95.20%	-0.029	0.318	1.190	0.00%	0.00%
DML-style IVQR	DGP1	1000	500	1	0.25	500	92.80%	-0.051	0.582	2.026	0.00%	0.00%
DML-style IVQR	DGP2	1000	500	1	0.25	500	97.00%	-0.042	0.637	2.298	0.20%	0.00%
DML-style IVQR	DGP3	1000	500	1	0.25	500	95.40%	-0.056	0.513	1.927	0.00%	0.00%
DML-style IVQR	DGP1	500	200	0.1	0.5	497	94.57%	0.046	1.282	3.740	72.40%	0.60%
DML-style IVQR	DGP2	500	200	0.1	0.5	500	95.00%	0.026	1.271	3.781	74.40%	0.00%
DML-style IVQR	DGP3	500	200	0.1	0.5	499	94.79%	0.048	1.240	3.759	72.60%	0.20%
DML-style IVQR	DGP1	500	500	0.1	0.5	499	94.19%	0.220	1.306	3.756	74.00%	0.20%
DML-style IVQR	DGP2	500	500	0.1	0.5	500	95.20%	0.076	1.328	3.781	74.60%	0.00%
DML-style IVQR	DGP3	500	500	0.1	0.5	500	95.20%	0.079	1.274	3.762	70.80%	0.00%
DML-style IVQR	DGP1	1000	200	0.1	0.5	500	95.20%	-0.100	1.216	3.693	61.00%	0.00%
DML-style IVQR	DGP2	1000	200	0.1	0.5	500	95.00%	0.026	1.226	3.710	67.00%	0.00%
DML-style IVQR	DGP3	1000	200	0.1	0.5	500	95.20%	0.034	1.169	3.609	53.60%	0.00%

Continued on next page

Table 10 (continued)

Estimator	DGP	n	p	pi	tau	R-res.	Coverage	Bias	RMSE	Mean CR	Full grid	Unresolved
DML-style IVQR	DGP1	1000	500	0.1	0.5	498	94.18%	0.058	1.326	3.767	71.60%	0.40%
DML-style IVQR	DGP2	1000	500	0.1	0.5	500	94.40%	0.081	1.330	3.761	75.20%	0.00%
DML-style IVQR	DGP3	1000	500	0.1	0.5	499	94.79%	0.078	1.258	3.703	67.60%	0.20%
DML-style IVQR	DGP1	500	200	0.25	0.5	499	94.99%	-0.030	1.135	3.575	54.00%	0.20%
DML-style IVQR	DGP2	500	200	0.25	0.5	500	94.60%	-0.046	1.174	3.613	52.40%	0.00%
DML-style IVQR	DGP3	500	200	0.25	0.5	500	94.60%	-0.022	1.094	3.438	38.40%	0.00%
DML-style IVQR	DGP1	500	500	0.25	0.5	500	94.60%	0.037	1.165	3.584	49.80%	0.00%
DML-style IVQR	DGP2	500	500	0.25	0.5	499	93.99%	-0.063	1.224	3.611	54.60%	0.20%
DML-style IVQR	DGP3	500	500	0.25	0.5	500	94.80%	-0.049	1.075	3.488	42.20%	0.00%
DML-style IVQR	DGP1	1000	200	0.25	0.5	500	95.00%	-0.134	0.860	2.999	13.20%	0.00%
DML-style IVQR	DGP2	1000	200	0.25	0.5	500	93.20%	-0.030	0.901	3.129	17.00%	0.00%
DML-style IVQR	DGP3	1000	200	0.25	0.5	500	94.20%	-0.064	0.643	2.628	4.00%	0.00%
DML-style IVQR	DGP1	1000	500	0.25	0.5	499	94.39%	-0.129	1.071	3.485	44.60%	0.20%
DML-style IVQR	DGP2	1000	500	0.25	0.5	500	94.80%	0.080	1.188	3.518	44.20%	0.00%
DML-style IVQR	DGP3	1000	500	0.25	0.5	500	96.20%	-0.016	0.974	3.320	30.80%	0.00%
DML-style IVQR	DGP1	500	200	0.5	0.5	500	95.20%	0.052	0.786	2.924	10.40%	0.00%
DML-style IVQR	DGP2	500	200	0.5	0.5	500	93.40%	-0.049	0.921	3.059	11.80%	0.00%
DML-style IVQR	DGP3	500	200	0.5	0.5	500	93.20%	-0.033	0.651	2.529	2.60%	0.00%
DML-style IVQR	DGP1	500	500	0.5	0.5	499	93.19%	-0.064	0.928	3.089	16.00%	0.20%
DML-style IVQR	DGP2	500	500	0.5	0.5	500	94.40%	-0.040	0.988	3.296	24.40%	0.00%
DML-style IVQR	DGP3	500	500	0.5	0.5	500	93.80%	-0.022	0.787	2.834	8.00%	0.00%
DML-style IVQR	DGP1	1000	200	0.5	0.5	500	96.20%	-0.044	0.422	1.761	0.00%	0.00%
DML-style IVQR	DGP2	1000	200	0.5	0.5	500	94.40%	-0.070	0.484	1.993	0.00%	0.00%
DML-style IVQR	DGP3	1000	200	0.5	0.5	500	94.40%	-0.023	0.326	1.385	0.00%	0.00%
DML-style IVQR	DGP1	1000	500	0.5	0.5	500	93.80%	-0.012	0.726	2.703	4.40%	0.00%
DML-style IVQR	DGP2	1000	500	0.5	0.5	500	94.60%	-0.053	0.847	2.918	8.80%	0.00%
DML-style IVQR	DGP3	1000	500	0.5	0.5	500	94.20%	-0.025	0.595	2.355	1.40%	0.00%
DML-style IVQR	DGP1	500	200	1	0.5	500	92.80%	-0.035	0.528	1.985	0.20%	0.00%
DML-style IVQR	DGP2	500	200	1	0.5	500	93.60%	0.022	0.546	2.160	0.00%	0.00%
DML-style IVQR	DGP3	500	200	1	0.5	500	93.20%	0.005	0.434	1.609	0.00%	0.00%
DML-style IVQR	DGP1	500	500	1	0.5	500	94.00%	-0.038	0.572	2.276	0.40%	0.00%
DML-style IVQR	DGP2	500	500	1	0.5	500	93.60%	0.010	0.664	2.432	0.60%	0.00%
DML-style IVQR	DGP3	500	500	1	0.5	500	94.60%	-0.036	0.470	1.879	0.00%	0.00%
DML-style IVQR	DGP1	1000	200	1	0.5	500	95.20%	-0.016	0.274	1.047	0.00%	0.00%
DML-style IVQR	DGP2	1000	200	1	0.5	500	95.60%	-0.055	0.292	1.143	0.00%	0.00%
DML-style IVQR	DGP3	1000	200	1	0.5	500	94.00%	-0.011	0.232	0.841	0.00%	0.00%
DML-style IVQR	DGP1	1000	500	1	0.5	500	94.00%	-0.019	0.442	1.759	0.00%	0.00%
DML-style IVQR	DGP2	1000	500	1	0.5	500	96.00%	-0.013	0.464	1.916	0.00%	0.00%
DML-style IVQR	DGP3	1000	500	1	0.5	500	95.20%	-0.035	0.367	1.429	0.00%	0.00%
DML-style IVQR	DGP1	500	200	0.1	0.75	499	95.59%	-0.275	1.231	3.780	79.80%	0.20%
DML-style IVQR	DGP2	500	200	0.1	0.75	500	94.00%	-0.350	1.352	3.754	77.40%	0.00%
DML-style IVQR	DGP3	500	200	0.1	0.75	499	97.19%	-0.186	1.218	3.836	79.40%	0.20%
DML-style IVQR	DGP1	500	500	0.1	0.75	500	93.80%	-0.302	1.269	3.760	76.80%	0.00%
DML-style IVQR	DGP2	500	500	0.1	0.75	499	94.79%	-0.392	1.323	3.791	79.60%	0.20%
DML-style IVQR	DGP3	500	500	0.1	0.75	500	96.00%	-0.190	1.262	3.788	76.20%	0.00%
DML-style IVQR	DGP1	1000	200	0.1	0.75	500	94.60%	-0.214	1.195	3.721	70.20%	0.00%
DML-style IVQR	DGP2	1000	200	0.1	0.75	499	96.19%	-0.297	1.286	3.783	77.20%	0.20%
DML-style IVQR	DGP3	1000	200	0.1	0.75	499	94.99%	-0.238	1.226	3.694	67.00%	0.20%
DML-style IVQR	DGP1	1000	500	0.1	0.75	500	96.00%	-0.257	1.242	3.801	78.00%	0.00%
DML-style IVQR	DGP2	1000	500	0.1	0.75	499	96.79%	-0.294	1.252	3.805	79.40%	0.20%
DML-style IVQR	DGP3	1000	500	0.1	0.75	500	96.60%	-0.206	1.249	3.749	72.80%	0.00%
DML-style IVQR	DGP1	500	200	0.25	0.75	500	96.40%	-0.240	1.226	3.616	61.20%	0.00%
DML-style IVQR	DGP2	500	200	0.25	0.75	500	95.00%	-0.265	1.213	3.664	66.40%	0.00%
DML-style IVQR	DGP3	500	200	0.25	0.75	500	94.80%	-0.245	1.081	3.619	57.20%	0.00%
DML-style IVQR	DGP1	500	500	0.25	0.75	500	93.00%	-0.200	1.219	3.609	65.40%	0.00%
DML-style IVQR	DGP2	500	500	0.25	0.75	499	94.39%	-0.352	1.299	3.650	65.40%	0.20%
DML-style IVQR	DGP3	500	500	0.25	0.75	499	95.79%	-0.188	1.183	3.635	60.20%	0.20%
DML-style IVQR	DGP1	1000	200	0.25	0.75	500	94.40%	-0.112	0.901	3.209	28.00%	0.00%
DML-style IVQR	DGP2	1000	200	0.25	0.75	500	94.40%	-0.201	0.948	3.394	39.00%	0.00%

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Table 10 (continued)

Estimator	DGP	n	p	pi	tau	R-res.	Coverage	Bias	RMSE	Mean CR	Full grid	Unresolved
DML-style IVQR	DGP3	1000	200	0.25	0.75	500	93.40%	-0.108	0.832	2.975	15.60%	0.00%
DML-style IVQR	DGP1	1000	500	0.25	0.75	498	95.18%	-0.217	1.158	3.558	56.60%	0.40%
DML-style IVQR	DGP2	1000	500	0.25	0.75	498	94.18%	-0.216	1.215	3.635	64.00%	0.40%
DML-style IVQR	DGP3	1000	500	0.25	0.75	499	94.19%	-0.183	1.160	3.500	51.00%	0.20%
DML-style IVQR	DGP1	500	200	0.5	0.75	500	95.00%	-0.188	0.937	3.178	25.20%	0.00%
DML-style IVQR	DGP2	500	200	0.5	0.75	500	93.20%	-0.193	1.032	3.230	32.00%	0.00%
DML-style IVQR	DGP3	500	200	0.5	0.75	500	96.20%	-0.074	0.801	2.994	16.20%	0.00%
DML-style IVQR	DGP1	500	500	0.5	0.75	499	95.99%	-0.157	0.990	3.311	38.20%	0.20%
DML-style IVQR	DGP2	500	500	0.5	0.75	500	95.60%	-0.190	1.022	3.398	42.00%	0.00%
DML-style IVQR	DGP3	500	500	0.5	0.75	500	94.80%	-0.090	0.919	3.096	20.20%	0.00%
DML-style IVQR	DGP1	1000	200	0.5	0.75	500	93.80%	-0.031	0.483	1.943	0.20%	0.00%
DML-style IVQR	DGP2	1000	200	0.5	0.75	500	95.20%	-0.082	0.550	2.248	1.20%	0.00%
DML-style IVQR	DGP3	1000	200	0.5	0.75	500	95.00%	-0.050	0.416	1.690	0.00%	0.00%
DML-style IVQR	DGP1	1000	500	0.5	0.75	500	91.80%	-0.081	0.839	2.929	19.40%	0.00%
DML-style IVQR	DGP2	1000	500	0.5	0.75	500	95.20%	-0.148	0.867	3.165	28.40%	0.00%
DML-style IVQR	DGP3	1000	500	0.5	0.75	500	95.80%	-0.124	0.725	2.788	9.40%	0.00%
DML-style IVQR	DGP1	500	200	1	0.75	500	94.40%	-0.057	0.556	2.170	0.40%	0.00%
DML-style IVQR	DGP2	500	200	1	0.75	500	93.20%	-0.051	0.634	2.316	1.80%	0.00%
DML-style IVQR	DGP3	500	200	1	0.75	500	94.80%	-0.050	0.486	1.926	0.60%	0.00%
DML-style IVQR	DGP1	500	500	1	0.75	500	95.20%	-0.052	0.621	2.470	3.00%	0.00%
DML-style IVQR	DGP2	500	500	1	0.75	500	95.60%	-0.144	0.719	2.674	7.20%	0.00%
DML-style IVQR	DGP3	500	500	1	0.75	500	94.80%	-0.056	0.591	2.202	0.60%	0.00%
DML-style IVQR	DGP1	1000	200	1	0.75	500	94.60%	-0.024	0.288	1.109	0.00%	0.00%
DML-style IVQR	DGP2	1000	200	1	0.75	500	94.60%	-0.047	0.310	1.243	0.00%	0.00%
DML-style IVQR	DGP3	1000	200	1	0.75	500	93.80%	-0.037	0.247	0.952	0.00%	0.00%
DML-style IVQR	DGP1	1000	500	1	0.75	500	94.20%	-0.049	0.474	1.900	0.20%	0.00%
DML-style IVQR	DGP2	1000	500	1	0.75	500	93.80%	-0.064	0.537	2.183	0.20%	0.00%
DML-style IVQR	DGP3	1000	500	1	0.75	500	96.00%	-0.026	0.397	1.683	0.20%	0.00%
Oracle IVQR	DGP1	500	200	0.1	0.25	500	95.20%	0.487	1.362	3.678	65.20%	0.00%
Oracle IVQR	DGP2	500	200	0.1	0.25	500	94.20%	0.495	1.379	3.663	65.60%	0.00%
Oracle IVQR	DGP3	500	200	0.1	0.25	500	95.40%	0.354	1.304	3.702	64.00%	0.00%
Oracle IVQR	DGP1	500	500	0.1	0.25	500	95.60%	0.541	1.398	3.708	64.60%	0.00%
Oracle IVQR	DGP2	500	500	0.1	0.25	500	95.60%	0.513	1.364	3.689	65.40%	0.00%
Oracle IVQR	DGP3	500	500	0.1	0.25	500	95.40%	0.277	1.244	3.627	56.40%	0.00%
Oracle IVQR	DGP1	1000	200	0.1	0.25	500	95.40%	0.442	1.274	3.610	56.20%	0.00%
Oracle IVQR	DGP2	1000	200	0.1	0.25	500	94.20%	0.467	1.325	3.615	57.20%	0.00%
Oracle IVQR	DGP3	1000	200	0.1	0.25	500	95.60%	0.248	1.210	3.558	50.80%	0.00%
Oracle IVQR	DGP1	1000	500	0.1	0.25	500	95.20%	0.334	1.248	3.616	58.40%	0.00%
Oracle IVQR	DGP2	1000	500	0.1	0.25	500	94.60%	0.377	1.268	3.618	57.60%	0.00%
Oracle IVQR	DGP3	1000	500	0.1	0.25	500	94.80%	0.233	1.174	3.552	49.20%	0.00%
Oracle IVQR	DGP1	500	200	0.25	0.25	500	95.00%	0.106	1.019	3.189	25.00%	0.00%
Oracle IVQR	DGP2	500	200	0.25	0.25	500	93.80%	0.112	1.069	3.252	26.80%	0.00%
Oracle IVQR	DGP3	500	200	0.25	0.25	500	95.40%	0.110	0.934	3.089	19.40%	0.00%
Oracle IVQR	DGP1	500	500	0.25	0.25	500	94.20%	0.104	1.021	3.179	25.60%	0.00%
Oracle IVQR	DGP2	500	500	0.25	0.25	500	94.00%	0.111	1.067	3.236	32.20%	0.00%
Oracle IVQR	DGP3	500	500	0.25	0.25	500	93.40%	0.030	0.896	2.973	14.00%	0.00%
Oracle IVQR	DGP1	1000	200	0.25	0.25	500	95.00%	0.012	0.840	2.749	6.20%	0.00%
Oracle IVQR	DGP2	1000	200	0.25	0.25	500	94.40%	0.112	0.842	2.906	11.00%	0.00%
Oracle IVQR	DGP3	1000	200	0.25	0.25	500	95.60%	-0.026	0.696	2.478	2.00%	0.00%
Oracle IVQR	DGP1	1000	500	0.25	0.25	500	95.20%	0.021	0.781	2.739	5.20%	0.00%
Oracle IVQR	DGP2	1000	500	0.25	0.25	500	95.40%	0.065	0.817	2.947	11.60%	0.00%
Oracle IVQR	DGP3	1000	500	0.25	0.25	500	93.80%	0.030	0.679	2.487	1.80%	0.00%
Oracle IVQR	DGP1	500	200	0.5	0.25	500	94.00%	-0.008	0.602	2.268	0.20%	0.00%
Oracle IVQR	DGP2	500	200	0.5	0.25	500	95.20%	0.027	0.663	2.390	1.40%	0.00%
Oracle IVQR	DGP3	500	200	0.5	0.25	500	94.80%	-0.077	0.533	2.004	0.00%	0.00%
Oracle IVQR	DGP1	500	500	0.5	0.25	500	94.80%	0.010	0.625	2.249	0.40%	0.00%
Oracle IVQR	DGP2	500	500	0.5	0.25	500	94.00%	-0.010	0.694	2.372	0.80%	0.00%
Oracle IVQR	DGP3	500	500	0.5	0.25	500	94.80%	-0.011	0.511	2.002	0.00%	0.00%
Oracle IVQR	DGP1	1000	200	0.5	0.25	500	93.80%	0.011	0.438	1.724	0.00%	0.00%

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Table 10 (continued)

Estimator	DGP	n	p	pi	tau	R-res.	Coverage	Bias	RMSE	Mean CR	Full grid	Unresolved
Oracle IVQR	DGP2	1000	200	0.5	0.25	500	92.80%	-0.007	0.487	1.828	0.00%	0.00%
Oracle IVQR	DGP3	1000	200	0.5	0.25	500	95.80%	0.008	0.377	1.463	0.00%	0.00%
Oracle IVQR	DGP1	1000	500	0.5	0.25	500	92.40%	0.000	0.486	1.700	0.00%	0.00%
Oracle IVQR	DGP2	1000	500	0.5	0.25	500	94.40%	0.020	0.485	1.840	0.00%	0.00%
Oracle IVQR	DGP3	1000	500	0.5	0.25	500	93.20%	-0.008	0.392	1.434	0.00%	0.00%
Oracle IVQR	DGP1	500	200	1	0.25	500	93.80%	-0.008	0.391	1.442	0.00%	0.00%
Oracle IVQR	DGP2	500	200	1	0.25	500	91.60%	-0.005	0.422	1.488	0.00%	0.00%
Oracle IVQR	DGP3	500	200	1	0.25	500	93.40%	-0.012	0.328	1.228	0.00%	0.00%
Oracle IVQR	DGP1	500	500	1	0.25	500	94.20%	-0.021	0.386	1.440	0.00%	0.00%
Oracle IVQR	DGP2	500	500	1	0.25	500	93.00%	0.012	0.412	1.529	0.00%	0.00%
Oracle IVQR	DGP3	500	500	1	0.25	500	93.40%	-0.013	0.332	1.216	0.00%	0.00%
Oracle IVQR	DGP1	1000	200	1	0.25	500	92.60%	-0.005	0.272	1.011	0.00%	0.00%
Oracle IVQR	DGP2	1000	200	1	0.25	500	92.60%	-0.011	0.295	1.082	0.00%	0.00%
Oracle IVQR	DGP3	1000	200	1	0.25	500	92.80%	0.017	0.226	0.838	0.00%	0.00%
Oracle IVQR	DGP1	1000	500	1	0.25	500	92.60%	0.004	0.274	1.006	0.00%	0.00%
Oracle IVQR	DGP2	1000	500	1	0.25	500	95.00%	0.013	0.272	1.068	0.00%	0.00%
Oracle IVQR	DGP3	1000	500	1	0.25	500	92.40%	0.018	0.231	0.844	0.00%	0.00%
Oracle IVQR	DGP1	500	200	0.1	0.5	500	95.20%	-0.039	1.253	3.747	67.40%	0.00%
Oracle IVQR	DGP2	500	200	0.1	0.5	500	95.60%	0.064	1.256	3.774	72.40%	0.00%
Oracle IVQR	DGP3	500	200	0.1	0.5	500	96.00%	0.018	1.246	3.702	62.60%	0.00%
Oracle IVQR	DGP1	500	500	0.1	0.5	500	94.80%	0.097	1.276	3.730	68.20%	0.00%
Oracle IVQR	DGP2	500	500	0.1	0.5	500	94.80%	0.050	1.310	3.726	68.80%	0.00%
Oracle IVQR	DGP3	500	500	0.1	0.5	500	95.60%	0.026	1.179	3.689	64.40%	0.00%
Oracle IVQR	DGP1	1000	200	0.1	0.5	500	94.40%	-0.052	1.204	3.606	54.20%	0.00%
Oracle IVQR	DGP2	1000	200	0.1	0.5	500	94.80%	0.080	1.250	3.662	60.40%	0.00%
Oracle IVQR	DGP3	1000	200	0.1	0.5	500	95.80%	0.015	1.110	3.551	47.00%	0.00%
Oracle IVQR	DGP1	1000	500	0.1	0.5	500	94.80%	0.131	1.236	3.614	57.00%	0.00%
Oracle IVQR	DGP2	1000	500	0.1	0.5	500	95.40%	-0.023	1.217	3.677	62.60%	0.00%
Oracle IVQR	DGP3	1000	500	0.1	0.5	500	95.60%	-0.048	1.071	3.573	45.40%	0.00%
Oracle IVQR	DGP1	500	200	0.25	0.5	500	94.80%	-0.058	1.013	3.270	23.00%	0.00%
Oracle IVQR	DGP2	500	200	0.25	0.5	500	94.60%	-0.065	1.055	3.352	29.00%	0.00%
Oracle IVQR	DGP3	500	200	0.25	0.5	500	97.40%	0.010	0.782	3.051	12.80%	0.00%
Oracle IVQR	DGP1	500	500	0.25	0.5	500	95.80%	0.007	0.938	3.319	22.80%	0.00%
Oracle IVQR	DGP2	500	500	0.25	0.5	500	95.40%	-0.059	1.012	3.392	34.00%	0.00%
Oracle IVQR	DGP3	500	500	0.25	0.5	500	96.00%	-0.083	0.789	3.029	13.80%	0.00%
Oracle IVQR	DGP1	1000	200	0.25	0.5	500	96.20%	-0.057	0.700	2.777	3.40%	0.00%
Oracle IVQR	DGP2	1000	200	0.25	0.5	500	93.80%	0.002	0.824	2.906	6.40%	0.00%
Oracle IVQR	DGP3	1000	200	0.25	0.5	500	95.40%	-0.018	0.543	2.356	0.40%	0.00%
Oracle IVQR	DGP1	1000	500	0.25	0.5	500	94.20%	-0.052	0.743	2.753	4.40%	0.00%
Oracle IVQR	DGP2	1000	500	0.25	0.5	500	95.80%	-0.029	0.791	2.947	8.60%	0.00%
Oracle IVQR	DGP3	1000	500	0.25	0.5	500	96.80%	-0.018	0.567	2.294	0.20%	0.00%
Oracle IVQR	DGP1	500	200	0.5	0.5	500	95.60%	-0.006	0.532	2.204	0.20%	0.00%
Oracle IVQR	DGP2	500	200	0.5	0.5	500	95.20%	-0.004	0.648	2.407	0.80%	0.00%
Oracle IVQR	DGP3	500	200	0.5	0.5	500	97.00%	0.013	0.410	1.791	0.00%	0.00%
Oracle IVQR	DGP1	500	500	0.5	0.5	500	96.20%	0.000	0.524	2.219	0.20%	0.00%
Oracle IVQR	DGP2	500	500	0.5	0.5	500	94.40%	-0.022	0.630	2.383	0.20%	0.00%
Oracle IVQR	DGP3	500	500	0.5	0.5	500	95.20%	-0.022	0.452	1.777	0.00%	0.00%
Oracle IVQR	DGP1	1000	200	0.5	0.5	500	96.60%	-0.021	0.364	1.511	0.00%	0.00%
Oracle IVQR	DGP2	1000	200	0.5	0.5	500	96.40%	-0.002	0.395	1.725	0.00%	0.00%
Oracle IVQR	DGP3	1000	200	0.5	0.5	500	94.40%	-0.008	0.276	1.175	0.00%	0.00%
Oracle IVQR	DGP1	1000	500	0.5	0.5	500	93.00%	-0.007	0.415	1.503	0.00%	0.00%
Oracle IVQR	DGP2	1000	500	0.5	0.5	500	93.80%	0.034	0.440	1.717	0.00%	0.00%
Oracle IVQR	DGP3	1000	500	0.5	0.5	500	96.20%	-0.009	0.296	1.182	0.00%	0.00%
Oracle IVQR	DGP1	500	200	1	0.5	500	94.80%	0.033	0.344	1.275	0.00%	0.00%
Oracle IVQR	DGP2	500	200	1	0.5	500	93.20%	0.034	0.373	1.367	0.00%	0.00%
Oracle IVQR	DGP3	500	200	1	0.5	500	95.00%	0.012	0.279	1.056	0.00%	0.00%
Oracle IVQR	DGP1	500	500	1	0.5	500	96.20%	-0.020	0.314	1.276	0.00%	0.00%
Oracle IVQR	DGP2	500	500	1	0.5	500	91.60%	-0.007	0.385	1.388	0.00%	0.00%
Oracle IVQR	DGP3	500	500	1	0.5	500	93.00%	-0.010	0.270	1.043	0.00%	0.00%

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Table 10 (continued)

Estimator	DGP	n	p	pi	tau	R-res.	Coverage	Bias	RMSE	Mean CR	Full grid	Unresolved
Oracle IVQR	DGP1	1000	200	1	0.5	500	93.80%	0.001	0.239	0.885	0.00%	0.00%
Oracle IVQR	DGP2	1000	200	1	0.5	500	93.20%	0.000	0.258	0.962	0.00%	0.00%
Oracle IVQR	DGP3	1000	200	1	0.5	500	92.80%	0.011	0.198	0.706	0.00%	0.00%
Oracle IVQR	DGP1	1000	500	1	0.5	500	93.00%	-0.000	0.242	0.892	0.00%	0.00%
Oracle IVQR	DGP2	1000	500	1	0.5	500	94.40%	0.005	0.243	0.956	0.00%	0.00%
Oracle IVQR	DGP3	1000	500	1	0.5	500	95.40%	-0.013	0.190	0.713	0.00%	0.00%
Oracle IVQR	DGP1	500	200	0.1	0.75	500	94.60%	-0.361	1.365	3.703	71.40%	0.00%
Oracle IVQR	DGP2	500	200	0.1	0.75	500	93.60%	-0.297	1.333	3.619	66.20%	0.00%
Oracle IVQR	DGP3	500	200	0.1	0.75	500	92.80%	-0.294	1.317	3.612	61.40%	0.00%
Oracle IVQR	DGP1	500	500	0.1	0.75	500	94.60%	-0.372	1.313	3.668	68.00%	0.00%
Oracle IVQR	DGP2	500	500	0.1	0.75	500	92.60%	-0.408	1.375	3.642	65.20%	0.00%
Oracle IVQR	DGP3	500	500	0.1	0.75	500	92.40%	-0.326	1.354	3.567	59.40%	0.00%
Oracle IVQR	DGP1	1000	200	0.1	0.75	500	96.00%	-0.385	1.315	3.633	60.80%	0.00%
Oracle IVQR	DGP2	1000	200	0.1	0.75	500	92.00%	-0.382	1.362	3.574	60.00%	0.00%
Oracle IVQR	DGP3	1000	200	0.1	0.75	500	94.20%	-0.298	1.253	3.577	54.00%	0.00%
Oracle IVQR	DGP1	1000	500	0.1	0.75	500	94.40%	-0.352	1.277	3.668	65.20%	0.00%
Oracle IVQR	DGP2	1000	500	0.1	0.75	500	95.20%	-0.429	1.377	3.697	63.80%	0.00%
Oracle IVQR	DGP3	1000	500	0.1	0.75	500	93.60%	-0.260	1.230	3.518	53.00%	0.00%
Oracle IVQR	DGP1	500	200	0.25	0.75	500	92.60%	-0.145	1.029	3.220	28.60%	0.00%
Oracle IVQR	DGP2	500	200	0.25	0.75	500	91.00%	-0.249	1.142	3.369	41.60%	0.00%
Oracle IVQR	DGP3	500	200	0.25	0.75	500	95.40%	-0.170	0.932	3.159	21.00%	0.00%
Oracle IVQR	DGP1	500	500	0.25	0.75	500	92.00%	-0.174	1.115	3.256	33.80%	0.00%
Oracle IVQR	DGP2	500	500	0.25	0.75	500	91.40%	-0.246	1.173	3.309	35.40%	0.00%
Oracle IVQR	DGP3	500	500	0.25	0.75	500	95.60%	-0.090	0.872	3.165	23.00%	0.00%
Oracle IVQR	DGP1	1000	200	0.25	0.75	500	93.40%	-0.070	0.844	2.868	10.80%	0.00%
Oracle IVQR	DGP2	1000	200	0.25	0.75	500	94.00%	-0.156	0.887	3.041	16.20%	0.00%
Oracle IVQR	DGP3	1000	200	0.25	0.75	500	91.80%	-0.058	0.751	2.572	4.40%	0.00%
Oracle IVQR	DGP1	1000	500	0.25	0.75	500	94.40%	-0.070	0.822	2.882	8.60%	0.00%
Oracle IVQR	DGP2	1000	500	0.25	0.75	500	91.60%	-0.164	0.937	3.024	19.00%	0.00%
Oracle IVQR	DGP3	1000	500	0.25	0.75	500	93.20%	-0.047	0.686	2.645	5.40%	0.00%
Oracle IVQR	DGP1	500	200	0.5	0.75	500	91.60%	-0.059	0.658	2.300	1.40%	0.00%
Oracle IVQR	DGP2	500	200	0.5	0.75	500	92.60%	-0.101	0.742	2.523	3.00%	0.00%
Oracle IVQR	DGP3	500	200	0.5	0.75	500	93.00%	-0.036	0.557	1.973	0.40%	0.00%
Oracle IVQR	DGP1	500	500	0.5	0.75	500	94.20%	-0.045	0.619	2.297	1.20%	0.00%
Oracle IVQR	DGP2	500	500	0.5	0.75	500	92.60%	-0.084	0.700	2.467	2.20%	0.00%
Oracle IVQR	DGP3	500	500	0.5	0.75	500	95.00%	-0.035	0.499	1.967	0.20%	0.00%
Oracle IVQR	DGP1	1000	200	0.5	0.75	500	92.40%	-0.015	0.460	1.643	0.00%	0.00%
Oracle IVQR	DGP2	1000	200	0.5	0.75	500	93.00%	-0.028	0.487	1.797	0.00%	0.00%
Oracle IVQR	DGP3	1000	200	0.5	0.75	500	94.40%	-0.008	0.353	1.377	0.00%	0.00%
Oracle IVQR	DGP1	1000	500	0.5	0.75	500	93.80%	-0.020	0.430	1.668	0.00%	0.00%
Oracle IVQR	DGP2	1000	500	0.5	0.75	500	93.00%	-0.019	0.490	1.848	0.20%	0.00%
Oracle IVQR	DGP3	1000	500	0.5	0.75	500	93.60%	-0.026	0.352	1.373	0.00%	0.00%
Oracle IVQR	DGP1	500	200	1	0.75	500	92.80%	-0.019	0.371	1.362	0.00%	0.00%
Oracle IVQR	DGP2	500	200	1	0.75	500	92.00%	0.001	0.398	1.433	0.00%	0.00%
Oracle IVQR	DGP3	500	200	1	0.75	500	92.00%	-0.018	0.316	1.146	0.00%	0.00%
Oracle IVQR	DGP1	500	500	1	0.75	500	93.80%	-0.005	0.331	1.347	0.00%	0.00%
Oracle IVQR	DGP2	500	500	1	0.75	500	94.60%	-0.018	0.377	1.453	0.00%	0.00%
Oracle IVQR	DGP3	500	500	1	0.75	500	91.20%	-0.016	0.322	1.150	0.00%	0.00%
Oracle IVQR	DGP1	1000	200	1	0.75	500	91.60%	0.018	0.259	0.945	0.00%	0.00%
Oracle IVQR	DGP2	1000	200	1	0.75	500	94.20%	-0.006	0.268	1.020	0.00%	0.00%
Oracle IVQR	DGP3	1000	200	1	0.75	500	94.00%	-0.015	0.219	0.792	0.00%	0.00%
Oracle IVQR	DGP1	1000	500	1	0.75	500	94.20%	-0.001	0.250	0.941	0.00%	0.00%
Oracle IVQR	DGP2	1000	500	1	0.75	500	92.80%	-0.010	0.284	1.001	0.00%	0.00%
Oracle IVQR	DGP3	1000	500	1	0.75	500	96.00%	0.009	0.209	0.788	0.00%	0.00%
Mean-Lasso PS-IVQR	DGP1	500	200	0.1	0.25	500	94.20%	0.549	1.404	3.634	63.60%	0.00%
Mean-Lasso PS-IVQR	DGP2	500	200	0.1	0.25	500	89.40%	0.482	1.423	3.525	59.40%	0.00%
Mean-Lasso PS-IVQR	DGP3	500	200	0.1	0.25	500	94.80%	0.429	1.332	3.658	61.00%	0.00%
Mean-Lasso PS-IVQR	DGP1	500	500	0.1	0.25	500	93.40%	0.532	1.394	3.632	60.60%	0.00%
Mean-Lasso PS-IVQR	DGP2	500	500	0.1	0.25	500	93.20%	0.425	1.328	3.545	57.60%	0.00%

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Table 10 (continued)

Estimator	DGP	n	p	pi	tau	R-res.	Coverage	Bias	RMSE	Mean CR	Full grid	Unresolved
Mean-Lasso PS-IVQR	DGP3	500	500	0.1	0.25	500	93.20%	0.296	1.271	3.560	52.80%	0.00%
Mean-Lasso PS-IVQR	DGP1	1000	200	0.1	0.25	500	96.40%	0.346	1.269	3.575	53.80%	0.00%
Mean-Lasso PS-IVQR	DGP2	1000	200	0.1	0.25	500	93.40%	0.471	1.335	3.556	53.00%	0.00%
Mean-Lasso PS-IVQR	DGP3	1000	200	0.1	0.25	500	94.60%	0.236	1.185	3.506	47.80%	0.00%
Mean-Lasso PS-IVQR	DGP1	1000	500	0.1	0.25	500	93.80%	0.390	1.283	3.572	52.80%	0.00%
Mean-Lasso PS-IVQR	DGP2	1000	500	0.1	0.25	500	93.60%	0.340	1.244	3.545	53.60%	0.00%
Mean-Lasso PS-IVQR	DGP3	1000	500	0.1	0.25	500	93.00%	0.289	1.173	3.502	45.40%	0.00%
Mean-Lasso PS-IVQR	DGP1	500	200	0.25	0.25	500	94.40%	0.094	1.000	3.146	23.60%	0.00%
Mean-Lasso PS-IVQR	DGP2	500	200	0.25	0.25	500	93.60%	0.204	1.088	3.167	26.40%	0.00%
Mean-Lasso PS-IVQR	DGP3	500	200	0.25	0.25	500	95.40%	0.107	0.955	3.032	16.60%	0.00%
Mean-Lasso PS-IVQR	DGP1	500	500	0.25	0.25	500	94.80%	0.096	1.009	3.192	23.40%	0.00%
Mean-Lasso PS-IVQR	DGP2	500	500	0.25	0.25	500	92.20%	0.071	1.115	3.130	28.20%	0.00%
Mean-Lasso PS-IVQR	DGP3	500	500	0.25	0.25	500	92.00%	0.000	0.956	2.878	13.00%	0.00%
Mean-Lasso PS-IVQR	DGP1	1000	200	0.25	0.25	500	92.80%	0.017	0.839	2.716	6.00%	0.00%
Mean-Lasso PS-IVQR	DGP2	1000	200	0.25	0.25	500	93.80%	0.093	0.864	2.836	9.60%	0.00%
Mean-Lasso PS-IVQR	DGP3	1000	200	0.25	0.25	500	95.40%	-0.026	0.669	2.435	1.80%	0.00%
Mean-Lasso PS-IVQR	DGP1	1000	500	0.25	0.25	500	95.20%	0.014	0.769	2.653	4.40%	0.00%
Mean-Lasso PS-IVQR	DGP2	1000	500	0.25	0.25	500	93.80%	0.043	0.862	2.826	8.40%	0.00%
Mean-Lasso PS-IVQR	DGP3	1000	500	0.25	0.25	500	93.00%	0.010	0.700	2.434	1.80%	0.00%
Mean-Lasso PS-IVQR	DGP1	500	200	0.5	0.25	500	94.20%	-0.026	0.617	2.219	0.20%	0.00%
Mean-Lasso PS-IVQR	DGP2	500	200	0.5	0.25	500	92.20%	0.038	0.697	2.314	1.20%	0.00%
Mean-Lasso PS-IVQR	DGP3	500	200	0.5	0.25	500	94.00%	-0.085	0.531	1.985	0.00%	0.00%
Mean-Lasso PS-IVQR	DGP1	500	500	0.5	0.25	500	93.40%	-0.018	0.637	2.200	0.60%	0.00%
Mean-Lasso PS-IVQR	DGP2	500	500	0.5	0.25	500	90.80%	-0.039	0.702	2.286	1.60%	0.00%
Mean-Lasso PS-IVQR	DGP3	500	500	0.5	0.25	500	93.40%	-0.054	0.524	1.934	0.00%	0.00%
Mean-Lasso PS-IVQR	DGP1	1000	200	0.5	0.25	500	93.40%	-0.009	0.441	1.683	0.00%	0.00%
Mean-Lasso PS-IVQR	DGP2	1000	200	0.5	0.25	500	93.20%	-0.001	0.492	1.762	0.00%	0.00%
Mean-Lasso PS-IVQR	DGP3	1000	200	0.5	0.25	500	94.60%	0.014	0.384	1.440	0.00%	0.00%
Mean-Lasso PS-IVQR	DGP1	1000	500	0.5	0.25	500	92.60%	0.007	0.480	1.661	0.00%	0.00%
Mean-Lasso PS-IVQR	DGP2	1000	500	0.5	0.25	500	93.80%	-0.008	0.493	1.805	0.00%	0.00%
Mean-Lasso PS-IVQR	DGP3	1000	500	0.5	0.25	500	90.00%	-0.016	0.420	1.400	0.00%	0.00%
Mean-Lasso PS-IVQR	DGP1	500	200	1	0.25	500	92.80%	-0.010	0.404	1.428	0.00%	0.00%
Mean-Lasso PS-IVQR	DGP2	500	200	1	0.25	500	89.40%	0.007	0.429	1.445	0.00%	0.00%
Mean-Lasso PS-IVQR	DGP3	500	200	1	0.25	500	92.00%	-0.009	0.332	1.212	0.00%	0.00%
Mean-Lasso PS-IVQR	DGP1	500	500	1	0.25	500	93.40%	-0.040	0.394	1.438	0.00%	0.00%
Mean-Lasso PS-IVQR	DGP2	500	500	1	0.25	500	90.60%	-0.002	0.434	1.463	0.00%	0.00%
Mean-Lasso PS-IVQR	DGP3	500	500	1	0.25	500	93.20%	-0.025	0.329	1.225	0.00%	0.00%
Mean-Lasso PS-IVQR	DGP1	1000	200	1	0.25	500	92.40%	-0.004	0.281	0.990	0.00%	0.00%
Mean-Lasso PS-IVQR	DGP2	1000	200	1	0.25	500	90.60%	0.004	0.299	1.040	0.00%	0.00%
Mean-Lasso PS-IVQR	DGP3	1000	200	1	0.25	500	93.00%	0.012	0.217	0.829	0.00%	0.00%
Mean-Lasso PS-IVQR	DGP1	1000	500	1	0.25	500	92.00%	-0.009	0.278	0.991	0.00%	0.00%
Mean-Lasso PS-IVQR	DGP2	1000	500	1	0.25	500	92.20%	-0.011	0.291	1.026	0.00%	0.00%
Mean-Lasso PS-IVQR	DGP3	1000	500	1	0.25	500	90.80%	0.013	0.248	0.846	0.00%	0.00%
Mean-Lasso PS-IVQR	DGP1	500	200	0.1	0.5	500	94.00%	-0.049	1.288	3.666	64.20%	0.00%
Mean-Lasso PS-IVQR	DGP2	500	200	0.1	0.5	500	92.80%	0.051	1.278	3.695	69.20%	0.00%
Mean-Lasso PS-IVQR	DGP3	500	200	0.1	0.5	500	94.80%	-0.075	1.238	3.662	59.60%	0.00%
Mean-Lasso PS-IVQR	DGP1	500	500	0.1	0.5	500	96.20%	0.058	1.289	3.711	69.40%	0.00%
Mean-Lasso PS-IVQR	DGP2	500	500	0.1	0.5	500	92.60%	0.092	1.316	3.627	63.40%	0.00%
Mean-Lasso PS-IVQR	DGP3	500	500	0.1	0.5	500	94.00%	0.043	1.222	3.663	60.80%	0.00%
Mean-Lasso PS-IVQR	DGP1	1000	200	0.1	0.5	500	94.40%	-0.071	1.192	3.559	50.00%	0.00%
Mean-Lasso PS-IVQR	DGP2	1000	200	0.1	0.5	499	93.19%	-0.006	1.275	3.593	55.40%	0.20%
Mean-Lasso PS-IVQR	DGP3	1000	200	0.1	0.5	500	95.80%	-0.017	1.112	3.500	43.60%	0.00%
Mean-Lasso PS-IVQR	DGP1	1000	500	0.1	0.5	500	94.20%	0.108	1.242	3.605	56.00%	0.00%
Mean-Lasso PS-IVQR	DGP2	1000	500	0.1	0.5	500	92.40%	0.016	1.254	3.580	55.20%	0.00%
Mean-Lasso PS-IVQR	DGP3	1000	500	0.1	0.5	500	96.00%	-0.044	1.077	3.521	44.20%	0.00%
Mean-Lasso PS-IVQR	DGP1	500	200	0.25	0.5	500	93.40%	-0.100	1.008	3.255	22.80%	0.00%
Mean-Lasso PS-IVQR	DGP2	500	200	0.25	0.5	500	92.80%	-0.105	1.101	3.264	26.00%	0.00%
Mean-Lasso PS-IVQR	DGP3	500	200	0.25	0.5	500	95.40%	-0.056	0.834	3.032	13.20%	0.00%
Mean-Lasso PS-IVQR	DGP1	500	500	0.25	0.5	500	94.60%	-0.078	0.975	3.258	21.80%	0.00%

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Table 10 (continued)

Estimator	DGP	n	p	pi	tau	R-res.	Coverage	Bias	RMSE	Mean CR	Full grid	Unresolved
Mean-Lasso PS-IVQR	DGP2	500	500	0.25	0.5	500	93.60%	-0.090	1.066	3.315	32.60%	0.00%
Mean-Lasso PS-IVQR	DGP3	500	500	0.25	0.5	500	94.60%	-0.198	0.861	3.009	11.60%	0.00%
Mean-Lasso PS-IVQR	DGP1	1000	200	0.25	0.5	500	94.60%	-0.115	0.730	2.752	4.40%	0.00%
Mean-Lasso PS-IVQR	DGP2	1000	200	0.25	0.5	500	93.60%	-0.048	0.841	2.880	7.40%	0.00%
Mean-Lasso PS-IVQR	DGP3	1000	200	0.25	0.5	500	95.60%	-0.081	0.552	2.347	1.60%	0.00%
Mean-Lasso PS-IVQR	DGP1	1000	500	0.25	0.5	500	95.00%	-0.056	0.752	2.747	5.20%	0.00%
Mean-Lasso PS-IVQR	DGP2	1000	500	0.25	0.5	500	93.20%	-0.094	0.813	2.891	7.80%	0.00%
Mean-Lasso PS-IVQR	DGP3	1000	500	0.25	0.5	500	95.60%	-0.032	0.587	2.321	0.60%	0.00%
Mean-Lasso PS-IVQR	DGP1	500	200	0.5	0.5	500	95.20%	-0.045	0.559	2.203	0.20%	0.00%
Mean-Lasso PS-IVQR	DGP2	500	200	0.5	0.5	500	94.40%	-0.088	0.664	2.348	0.80%	0.00%
Mean-Lasso PS-IVQR	DGP3	500	200	0.5	0.5	500	96.60%	-0.045	0.431	1.833	0.00%	0.00%
Mean-Lasso PS-IVQR	DGP1	500	500	0.5	0.5	500	95.00%	-0.108	0.617	2.225	0.00%	0.00%
Mean-Lasso PS-IVQR	DGP2	500	500	0.5	0.5	500	92.40%	-0.177	0.697	2.344	1.00%	0.00%
Mean-Lasso PS-IVQR	DGP3	500	500	0.5	0.5	500	95.00%	-0.094	0.480	1.815	0.00%	0.00%
Mean-Lasso PS-IVQR	DGP1	1000	200	0.5	0.5	500	95.00%	-0.055	0.367	1.509	0.00%	0.00%
Mean-Lasso PS-IVQR	DGP2	1000	200	0.5	0.5	500	94.20%	-0.039	0.438	1.678	0.00%	0.00%
Mean-Lasso PS-IVQR	DGP3	1000	200	0.5	0.5	500	95.20%	-0.031	0.283	1.190	0.00%	0.00%
Mean-Lasso PS-IVQR	DGP1	1000	500	0.5	0.5	500	92.20%	-0.067	0.427	1.529	0.00%	0.00%
Mean-Lasso PS-IVQR	DGP2	1000	500	0.5	0.5	500	92.60%	-0.042	0.459	1.698	0.00%	0.00%
Mean-Lasso PS-IVQR	DGP3	1000	500	0.5	0.5	500	94.20%	-0.042	0.314	1.208	0.00%	0.00%
Mean-Lasso PS-IVQR	DGP1	500	200	1	0.5	500	92.20%	-0.033	0.361	1.281	0.00%	0.00%
Mean-Lasso PS-IVQR	DGP2	500	200	1	0.5	500	90.00%	-0.025	0.383	1.343	0.00%	0.00%
Mean-Lasso PS-IVQR	DGP3	500	200	1	0.5	500	93.40%	-0.032	0.285	1.065	0.00%	0.00%
Mean-Lasso PS-IVQR	DGP1	500	500	1	0.5	500	93.40%	-0.103	0.362	1.306	0.00%	0.00%
Mean-Lasso PS-IVQR	DGP2	500	500	1	0.5	500	90.20%	-0.104	0.414	1.404	0.00%	0.00%
Mean-Lasso PS-IVQR	DGP3	500	500	1	0.5	500	90.20%	-0.070	0.287	1.073	0.00%	0.00%
Mean-Lasso PS-IVQR	DGP1	1000	200	1	0.5	500	92.00%	-0.020	0.247	0.895	0.00%	0.00%
Mean-Lasso PS-IVQR	DGP2	1000	200	1	0.5	500	93.60%	-0.036	0.261	0.947	0.00%	0.00%
Mean-Lasso PS-IVQR	DGP3	1000	200	1	0.5	500	92.00%	-0.010	0.208	0.706	0.00%	0.00%
Mean-Lasso PS-IVQR	DGP1	1000	500	1	0.5	500	93.40%	-0.042	0.251	0.891	0.00%	0.00%
Mean-Lasso PS-IVQR	DGP2	1000	500	1	0.5	500	92.40%	-0.051	0.257	0.951	0.00%	0.00%
Mean-Lasso PS-IVQR	DGP3	1000	500	1	0.5	500	93.20%	-0.046	0.205	0.714	0.00%	0.00%
Mean-Lasso PS-IVQR	DGP1	500	200	0.1	0.75	500	92.40%	-0.348	1.336	3.660	68.80%	0.00%
Mean-Lasso PS-IVQR	DGP2	500	200	0.1	0.75	500	92.80%	-0.357	1.334	3.575	63.40%	0.00%
Mean-Lasso PS-IVQR	DGP3	500	200	0.1	0.75	500	92.80%	-0.335	1.311	3.597	60.20%	0.00%
Mean-Lasso PS-IVQR	DGP1	500	500	0.1	0.75	500	91.60%	-0.434	1.347	3.595	62.60%	0.00%
Mean-Lasso PS-IVQR	DGP2	500	500	0.1	0.75	500	89.60%	-0.522	1.429	3.500	59.00%	0.00%
Mean-Lasso PS-IVQR	DGP3	500	500	0.1	0.75	500	91.60%	-0.364	1.367	3.513	57.20%	0.00%
Mean-Lasso PS-IVQR	DGP1	1000	200	0.1	0.75	500	92.60%	-0.340	1.310	3.575	58.20%	0.00%
Mean-Lasso PS-IVQR	DGP2	1000	200	0.1	0.75	500	88.40%	-0.402	1.327	3.560	59.00%	0.00%
Mean-Lasso PS-IVQR	DGP3	1000	200	0.1	0.75	500	92.60%	-0.260	1.241	3.520	49.40%	0.00%
Mean-Lasso PS-IVQR	DGP1	1000	500	0.1	0.75	500	94.00%	-0.388	1.327	3.622	60.40%	0.00%
Mean-Lasso PS-IVQR	DGP2	1000	500	0.1	0.75	499	93.39%	-0.473	1.387	3.591	59.00%	0.20%
Mean-Lasso PS-IVQR	DGP3	1000	500	0.1	0.75	500	94.00%	-0.254	1.218	3.487	50.20%	0.00%
Mean-Lasso PS-IVQR	DGP1	500	200	0.25	0.75	500	92.80%	-0.306	1.109	3.250	27.40%	0.00%
Mean-Lasso PS-IVQR	DGP2	500	200	0.25	0.75	500	90.20%	-0.320	1.205	3.265	34.20%	0.00%
Mean-Lasso PS-IVQR	DGP3	500	200	0.25	0.75	500	92.20%	-0.267	0.980	3.091	19.60%	0.00%
Mean-Lasso PS-IVQR	DGP1	500	500	0.25	0.75	500	90.00%	-0.335	1.171	3.274	29.80%	0.00%
Mean-Lasso PS-IVQR	DGP2	500	500	0.25	0.75	500	86.80%	-0.472	1.279	3.244	35.00%	0.00%
Mean-Lasso PS-IVQR	DGP3	500	500	0.25	0.75	500	91.00%	-0.219	1.012	3.198	23.40%	0.00%
Mean-Lasso PS-IVQR	DGP1	1000	200	0.25	0.75	500	91.40%	-0.131	0.898	2.850	9.80%	0.00%
Mean-Lasso PS-IVQR	DGP2	1000	200	0.25	0.75	500	91.80%	-0.254	0.952	3.010	17.20%	0.00%
Mean-Lasso PS-IVQR	DGP3	1000	200	0.25	0.75	500	90.00%	-0.091	0.765	2.570	3.60%	0.00%
Mean-Lasso PS-IVQR	DGP1	1000	500	0.25	0.75	500	92.00%	-0.181	0.870	2.858	10.20%	0.00%
Mean-Lasso PS-IVQR	DGP2	1000	500	0.25	0.75	500	90.40%	-0.296	1.042	2.992	14.60%	0.00%
Mean-Lasso PS-IVQR	DGP3	1000	500	0.25	0.75	500	92.20%	-0.185	0.775	2.679	5.80%	0.00%
Mean-Lasso PS-IVQR	DGP1	500	200	0.5	0.75	500	89.60%	-0.168	0.717	2.384	1.40%	0.00%
Mean-Lasso PS-IVQR	DGP2	500	200	0.5	0.75	500	88.00%	-0.270	0.825	2.513	3.60%	0.00%
Mean-Lasso PS-IVQR	DGP3	500	200	0.5	0.75	500	90.00%	-0.166	0.641	2.023	0.20%	0.00%

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Table 10 (continued)

Estimator	DGP	n	p	pi	tau	R-res.	Coverage	Bias	RMSE	Mean CR	Full grid	Unresolved
Mean-Lasso PS-IVQR	DGP1	500	500	0.5	0.75	500	90.60%	-0.231	0.726	2.392	1.40%	0.00%
Mean-Lasso PS-IVQR	DGP2	500	500	0.5	0.75	500	87.00%	-0.352	0.840	2.502	2.00%	0.00%
Mean-Lasso PS-IVQR	DGP3	500	500	0.5	0.75	500	91.60%	-0.203	0.620	2.064	0.20%	0.00%
Mean-Lasso PS-IVQR	DGP1	1000	200	0.5	0.75	500	89.80%	-0.093	0.483	1.658	0.00%	0.00%
Mean-Lasso PS-IVQR	DGP2	1000	200	0.5	0.75	500	89.40%	-0.102	0.529	1.779	0.00%	0.00%
Mean-Lasso PS-IVQR	DGP3	1000	200	0.5	0.75	500	90.60%	-0.070	0.381	1.374	0.00%	0.00%
Mean-Lasso PS-IVQR	DGP1	1000	500	0.5	0.75	500	92.60%	-0.114	0.470	1.689	0.00%	0.00%
Mean-Lasso PS-IVQR	DGP2	1000	500	0.5	0.75	500	88.00%	-0.154	0.570	1.836	0.00%	0.00%
Mean-Lasso PS-IVQR	DGP3	1000	500	0.5	0.75	500	91.00%	-0.132	0.404	1.406	0.00%	0.00%
Mean-Lasso PS-IVQR	DGP1	500	200	1	0.75	500	90.40%	-0.152	0.423	1.376	0.00%	0.00%
Mean-Lasso PS-IVQR	DGP2	500	200	1	0.75	500	85.80%	-0.146	0.452	1.452	0.00%	0.00%
Mean-Lasso PS-IVQR	DGP3	500	200	1	0.75	500	90.00%	-0.114	0.365	1.172	0.00%	0.00%
Mean-Lasso PS-IVQR	DGP1	500	500	1	0.75	500	90.80%	-0.190	0.437	1.420	0.00%	0.00%
Mean-Lasso PS-IVQR	DGP2	500	500	1	0.75	500	83.80%	-0.256	0.516	1.478	0.00%	0.00%
Mean-Lasso PS-IVQR	DGP3	500	500	1	0.75	500	83.80%	-0.157	0.391	1.171	0.00%	0.00%
Mean-Lasso PS-IVQR	DGP1	1000	200	1	0.75	500	90.00%	-0.038	0.273	0.934	0.00%	0.00%
Mean-Lasso PS-IVQR	DGP2	1000	200	1	0.75	500	90.20%	-0.078	0.295	0.995	0.00%	0.00%
Mean-Lasso PS-IVQR	DGP3	1000	200	1	0.75	500	93.40%	-0.066	0.233	0.795	0.00%	0.00%
Mean-Lasso PS-IVQR	DGP1	1000	500	1	0.75	500	88.20%	-0.086	0.290	0.940	0.00%	0.00%
Mean-Lasso PS-IVQR	DGP2	1000	500	1	0.75	500	88.00%	-0.107	0.339	1.008	0.00%	0.00%
Mean-Lasso PS-IVQR	DGP3	1000	500	1	0.75	500	90.80%	-0.064	0.231	0.797	0.00%	0.00%

References

Chen, Jau-er, Chien-Hsun Huang, and Jia-Jyun Tien. 2021. "Debiased/Double Machine Learning for Instrumental Variable Quantile Regressions." *Econometrics* 9 (2): 15. https://doi.org/10.3390/econometrics9020015.

Chernozhukov, Victor, Denis Chetverikov, Mert Demirer, Esther Duflo, Christian Hansen, Whitney Newey, and James Robins. 2018. "Double/Debiased Machine Learning for Treatment and Structural Parameters." *The Econometrics Journal* 21 (1): C1–68. https://doi.org/10.1111/ectj.12097.

Chernozhukov, Victor, and Christian Hansen. 2005. "An IV Model of Quantile Treatment Effects." *Econometrica* 73 (1): 245–61. https://doi.org/10.1111/j.1468-0262.2005.00570.x.

———. 2006. "Instrumental Quantile Regression Inference for Structural and Treatment Effect Models." *Journal of Econometrics* 132 (2): 491–525. https://doi.org/10.1016/j.jeconom.2005.02.009.

———. 2008. "Instrumental Variable Quantile Regression: A Robust Inference Approach." *Journal of Econometrics* 142 (1): 379–98. https://doi.org/10.1016/j.jeconom.2007.06.005.

Chernozhukov, Victor, Christian Hansen, and Michael Jansson. 2009. "Finite Sample Inference for Quantile Regression Models." *Journal of Econometrics* 152 (2): 93–103. https://doi.org/10.1016/j.jeconom.2009.01.004.

Chernozhukov, Victor, Christian Hansen, and Martin Spindler. 2015. "Valid Post-Selection and Post-Regularization Inference: An Elementary, General Approach." *Annual Review of Economics* 7: 649–88. https://doi.org/10.1146/annurev-economics-012315-015826.

Jun, Sung Jae. 2008. "Weak Identification Robust Tests in an Instrumental Quantile Model." *Journal of Econometrics* 144 (1): 118–38. https://doi.org/10.1016/j.jeconom.2007.12.006.